

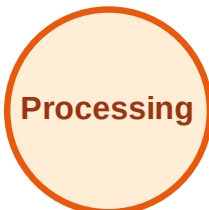
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



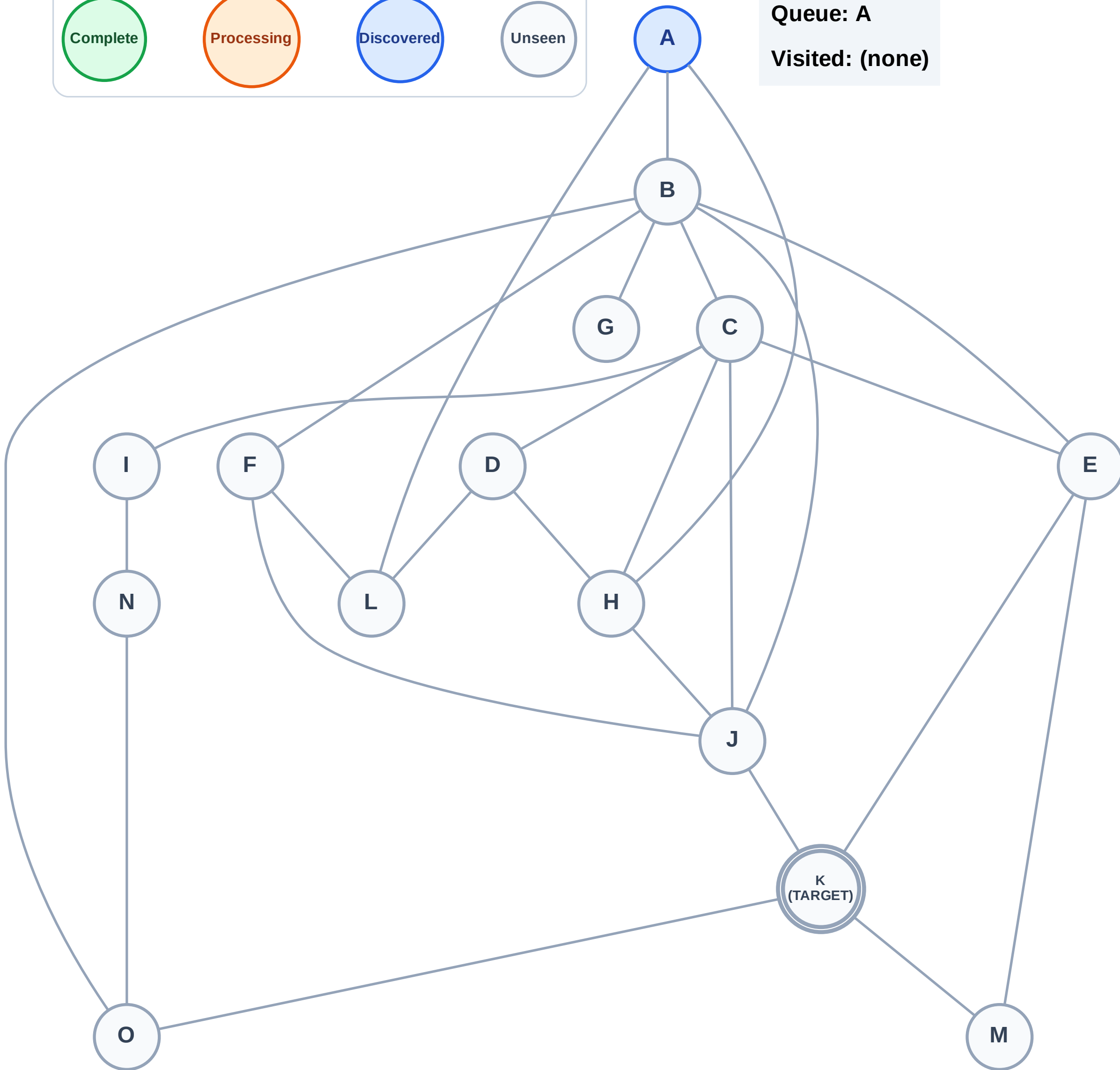
Discovered



Unseen

Queue: A

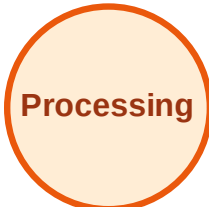
Visited: (none)



BFS Traversal

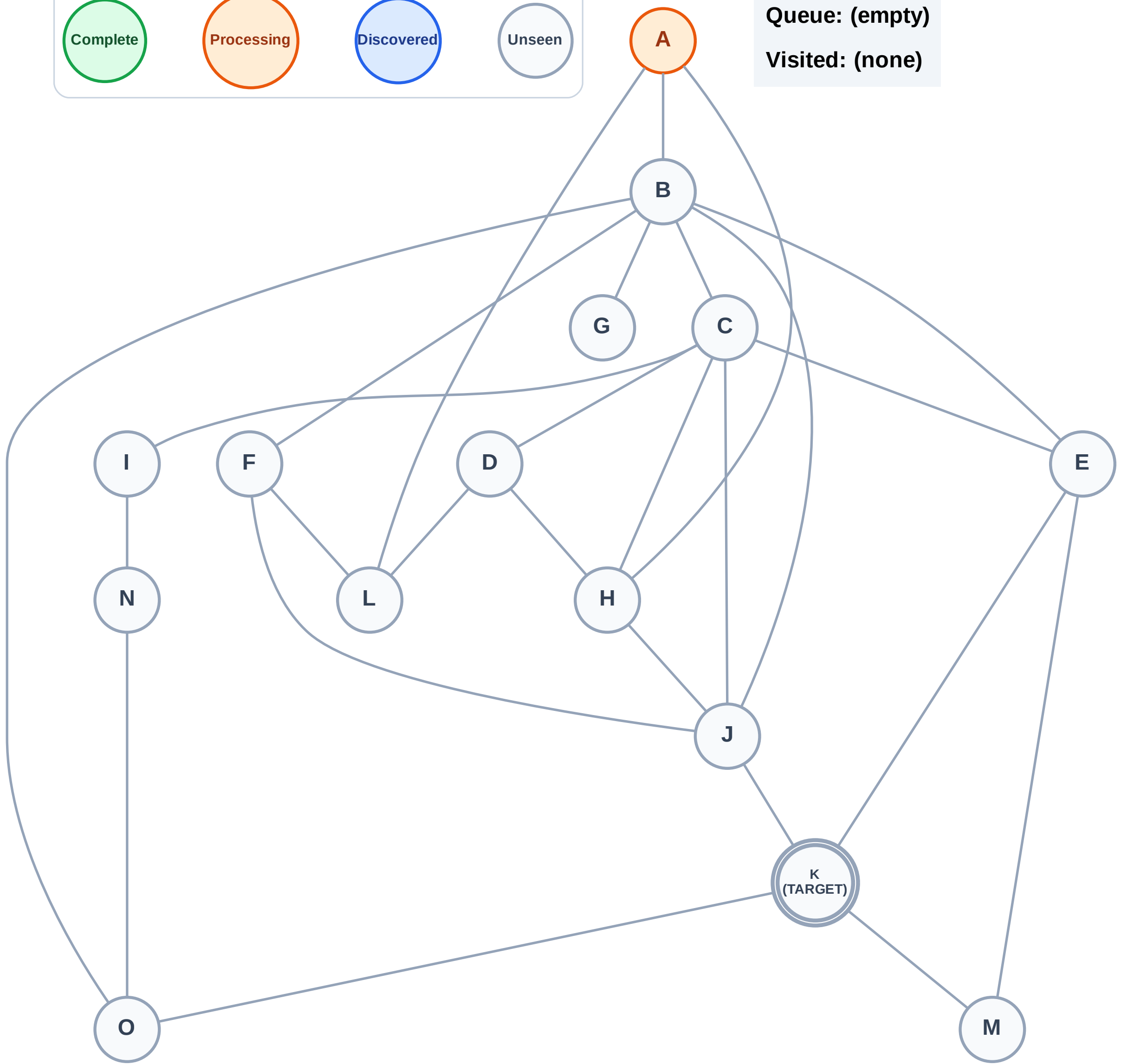
Step 2: Process node A

Legend



Queue: (empty)

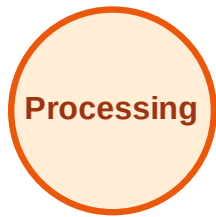
Visited: (none)



BFS Traversal

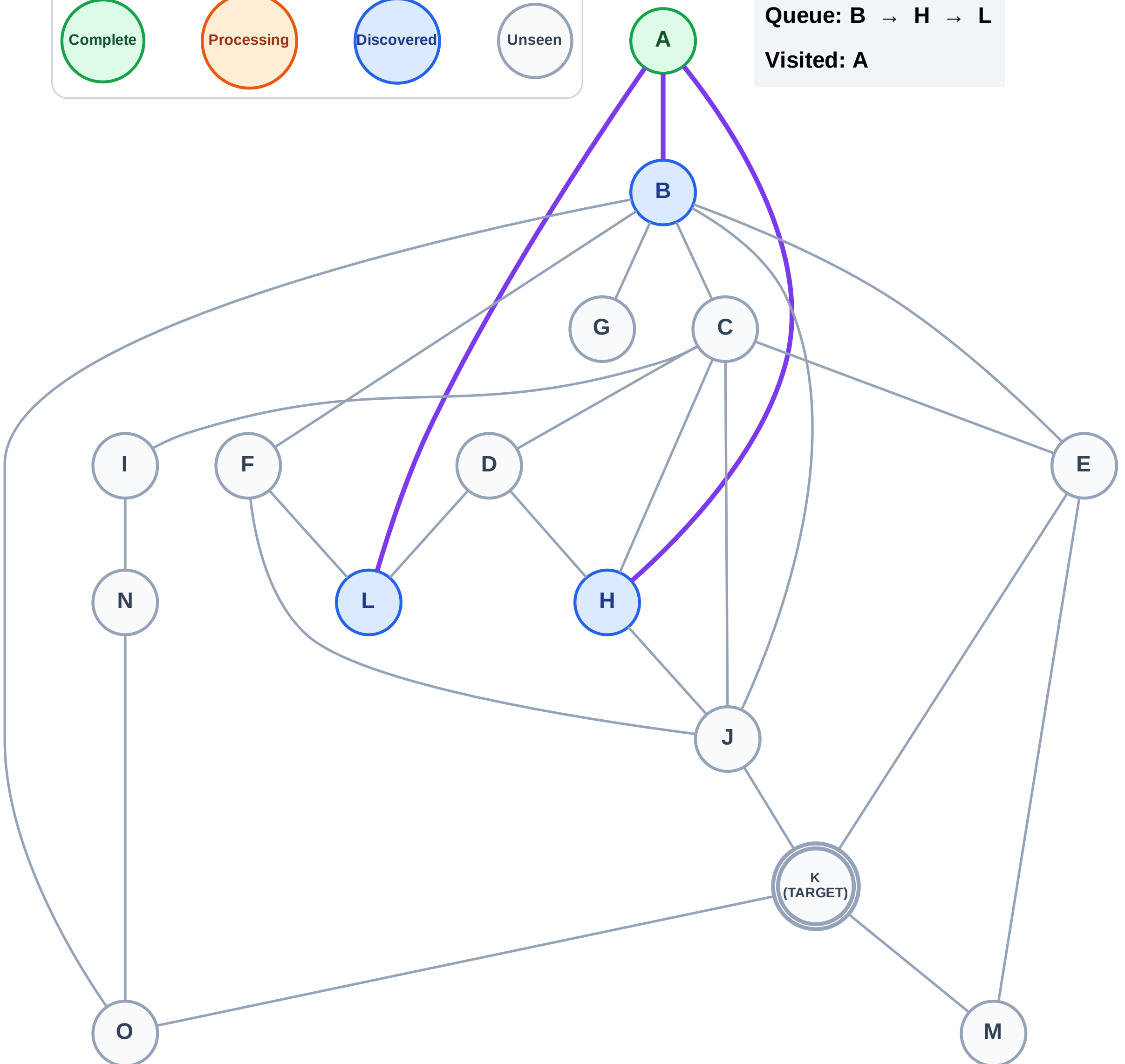
Step 3: Discover B, H, L from A

Legend



Queue: B → H → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

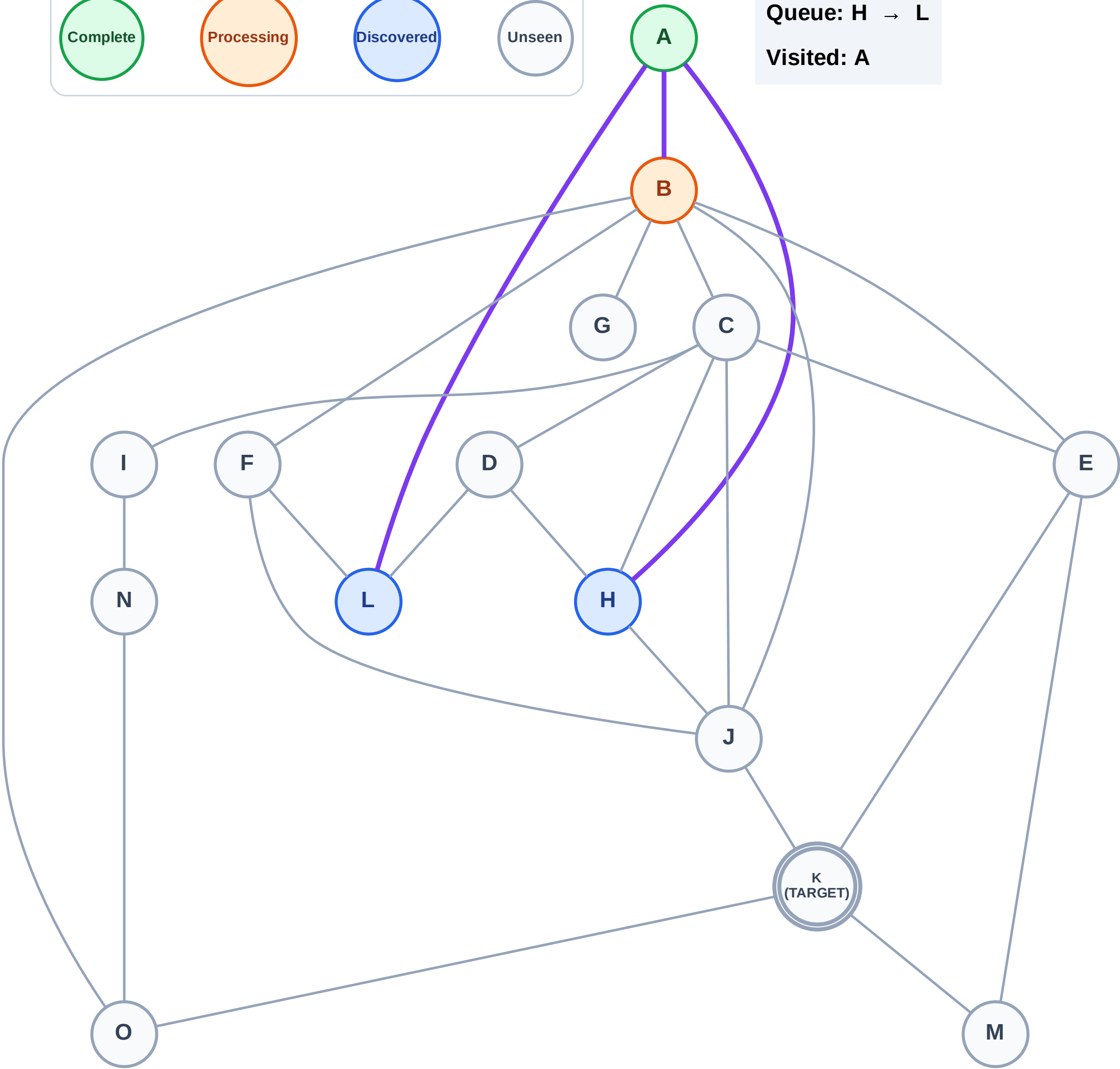
Processing

Discovered

Unseen

Queue: H → L

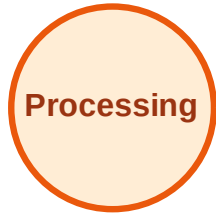
Visited: A



BFS Traversal

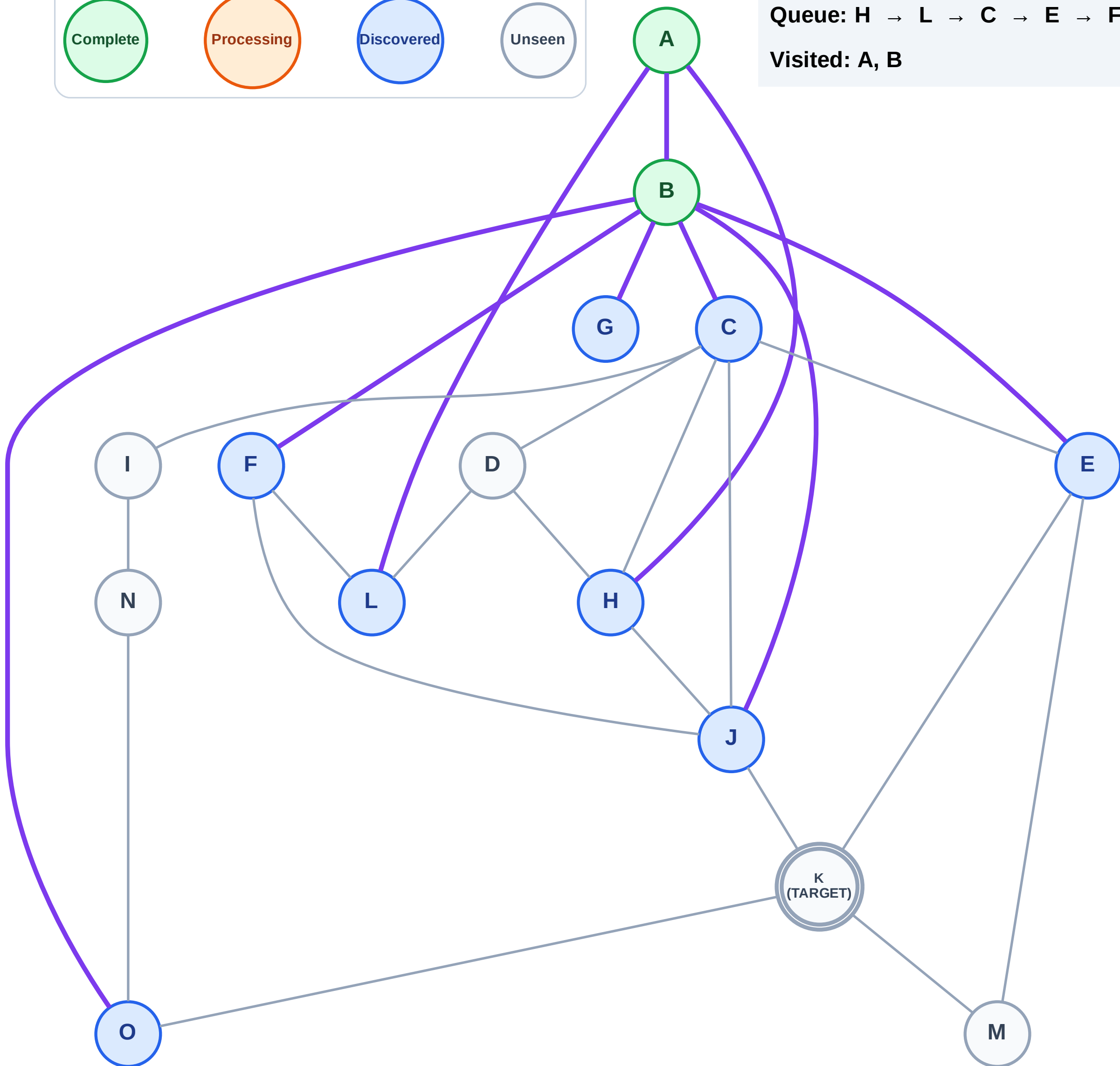
Step 5: Discover C, E, F, G, J, O from B

Legend



Queue: H → L → C → E → F → G → J → O

Visited: A, B



BFS Traversal

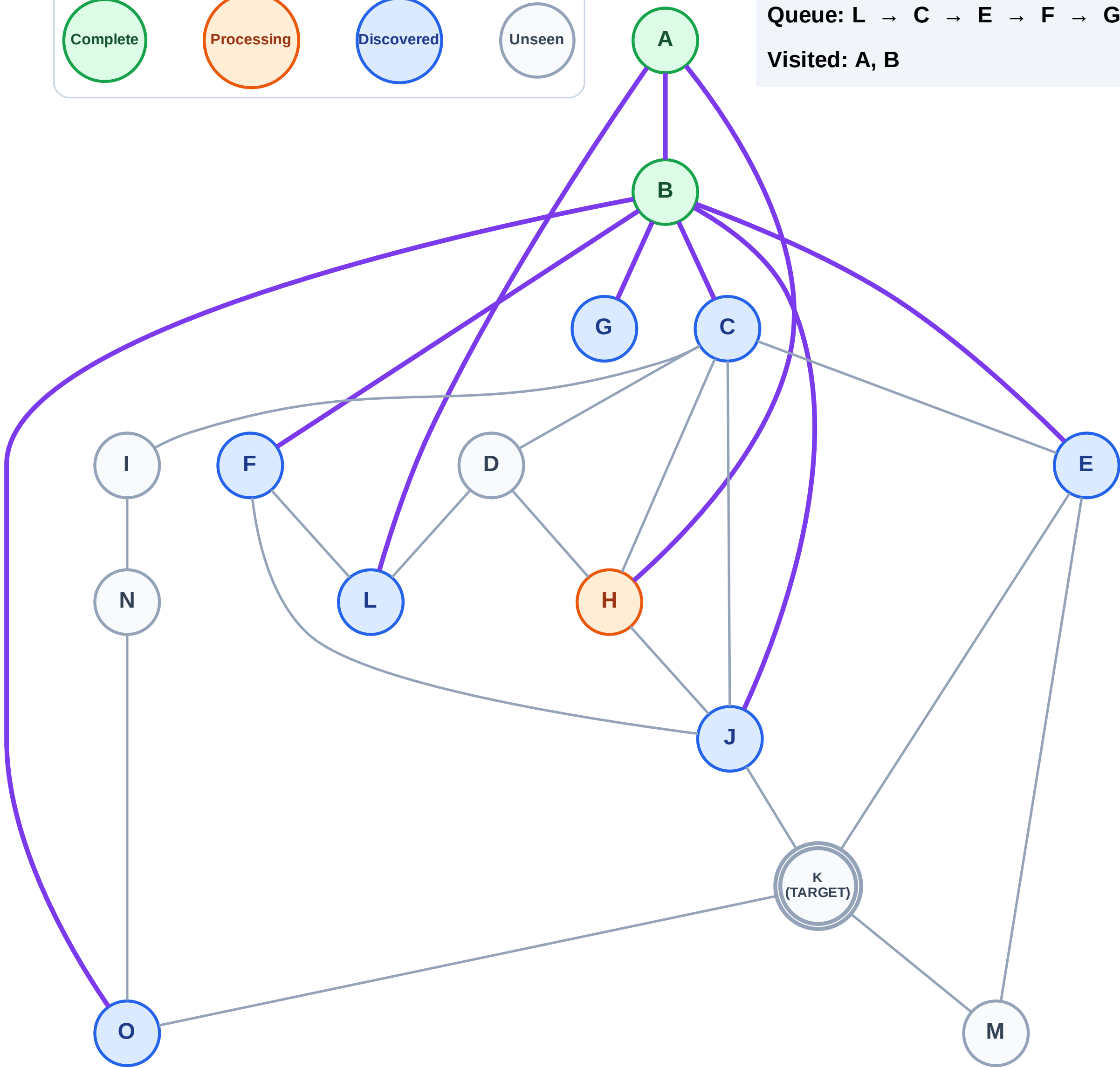
Step 6: Process node H

Legend



Queue: L → C → E → F → G → J → O

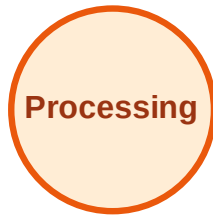
Visited: A, B



BFS Traversal

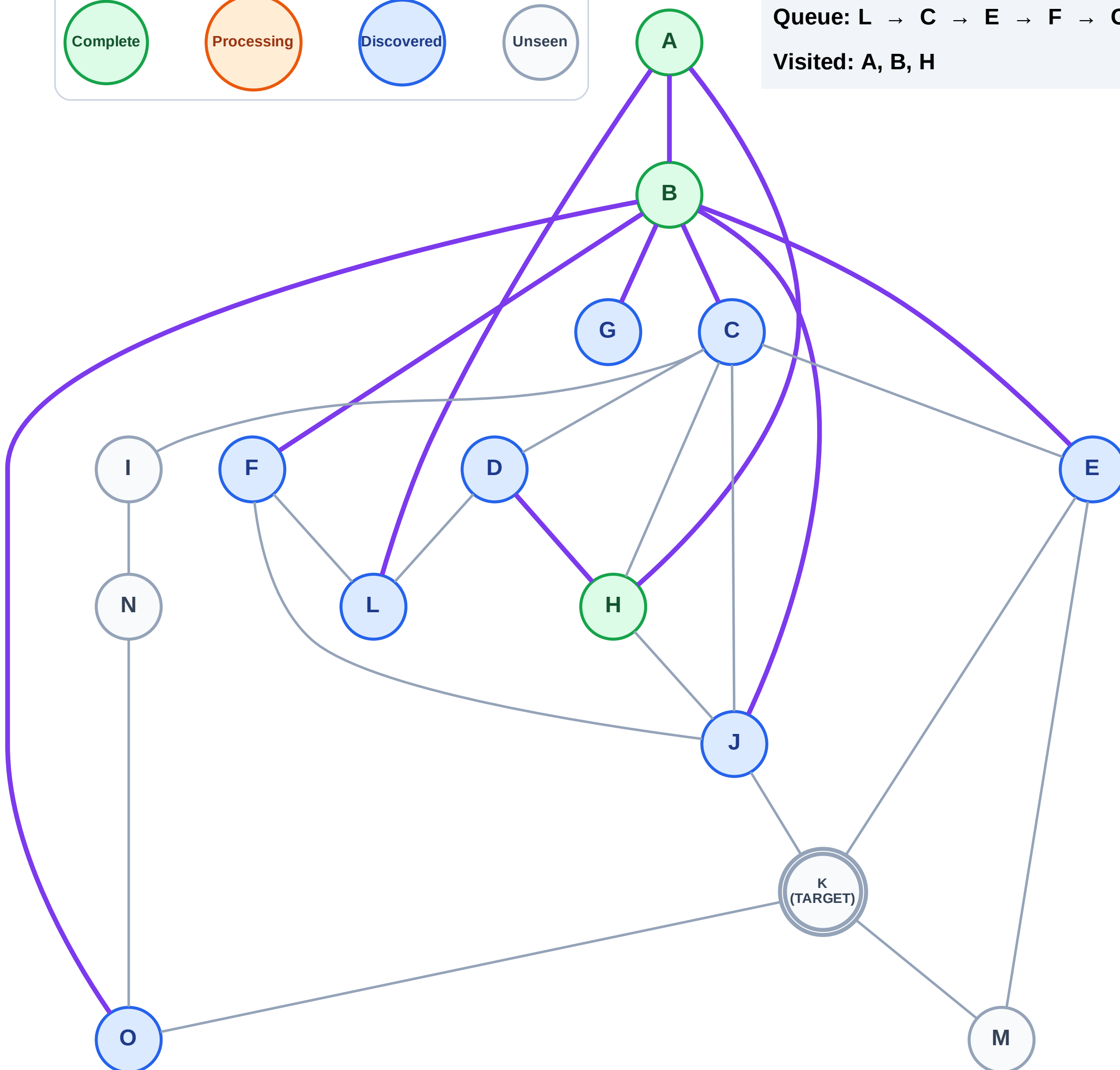
Step 7: Discover D from H

Legend



Queue: L → C → E → F → G → J → O → D

Visited: A, B, H



BFS Traversal

Step 8: Process node L

Legend

Complete

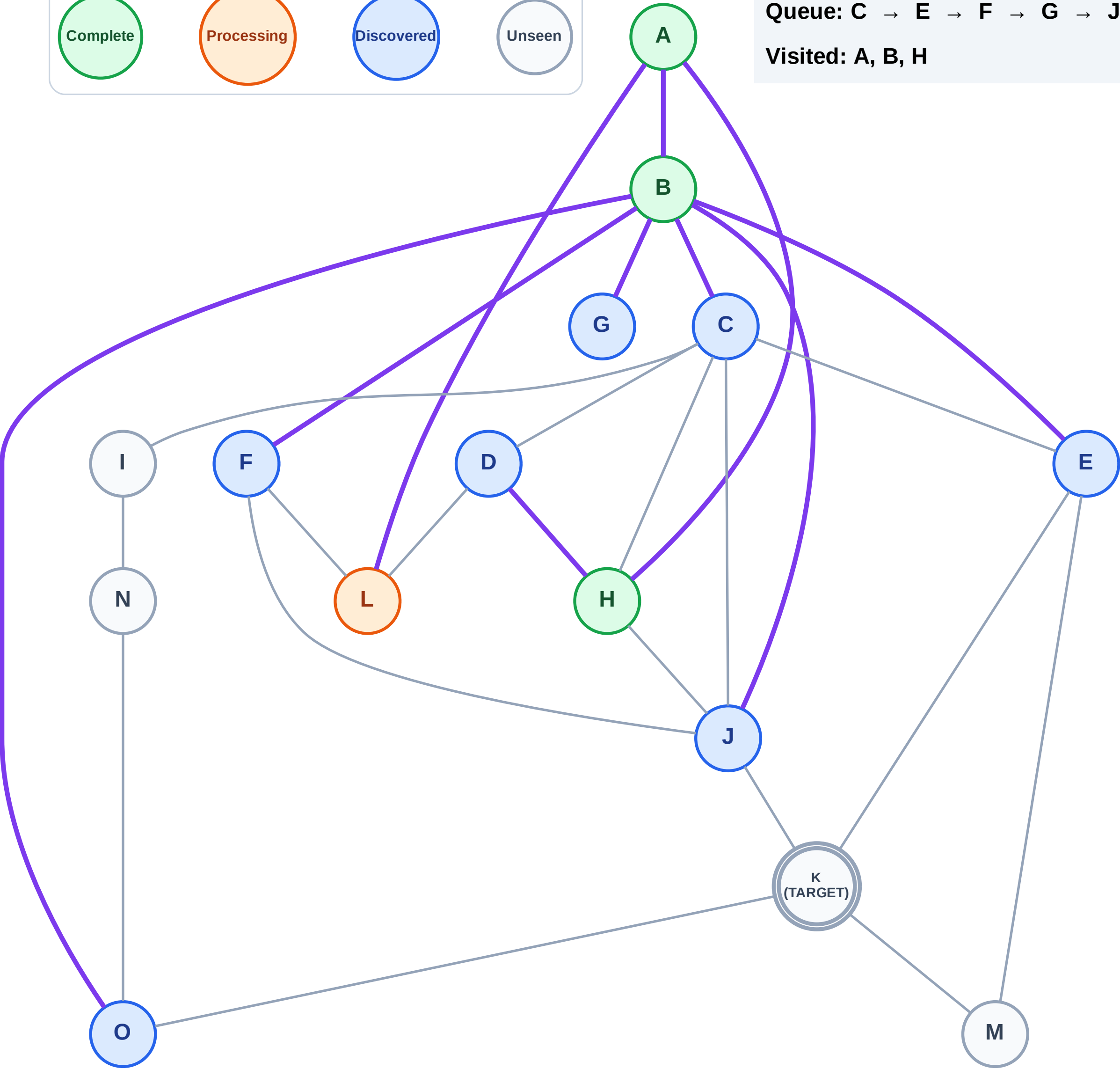
Processing

Discovered

Unseen

Queue: C → E → F → G → J → O → D

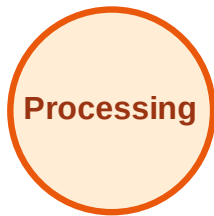
Visited: A, B, H



BFS Traversal

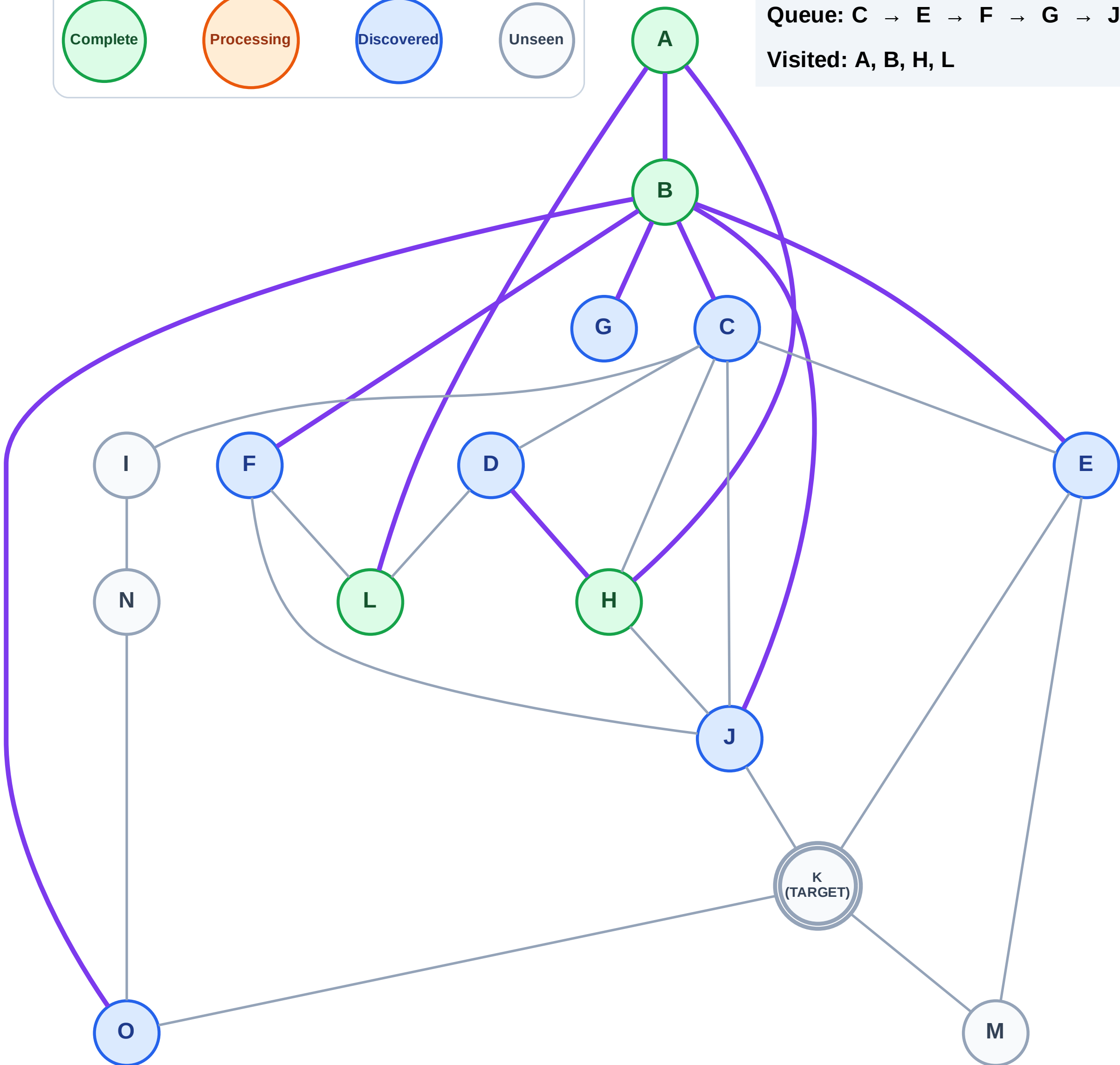
Step 9: No new neighbors from L

Legend



Queue: C → E → F → G → J → O → D

Visited: A, B, H, L



BFS Traversal

Step 10: Process node C

Legend

Complete

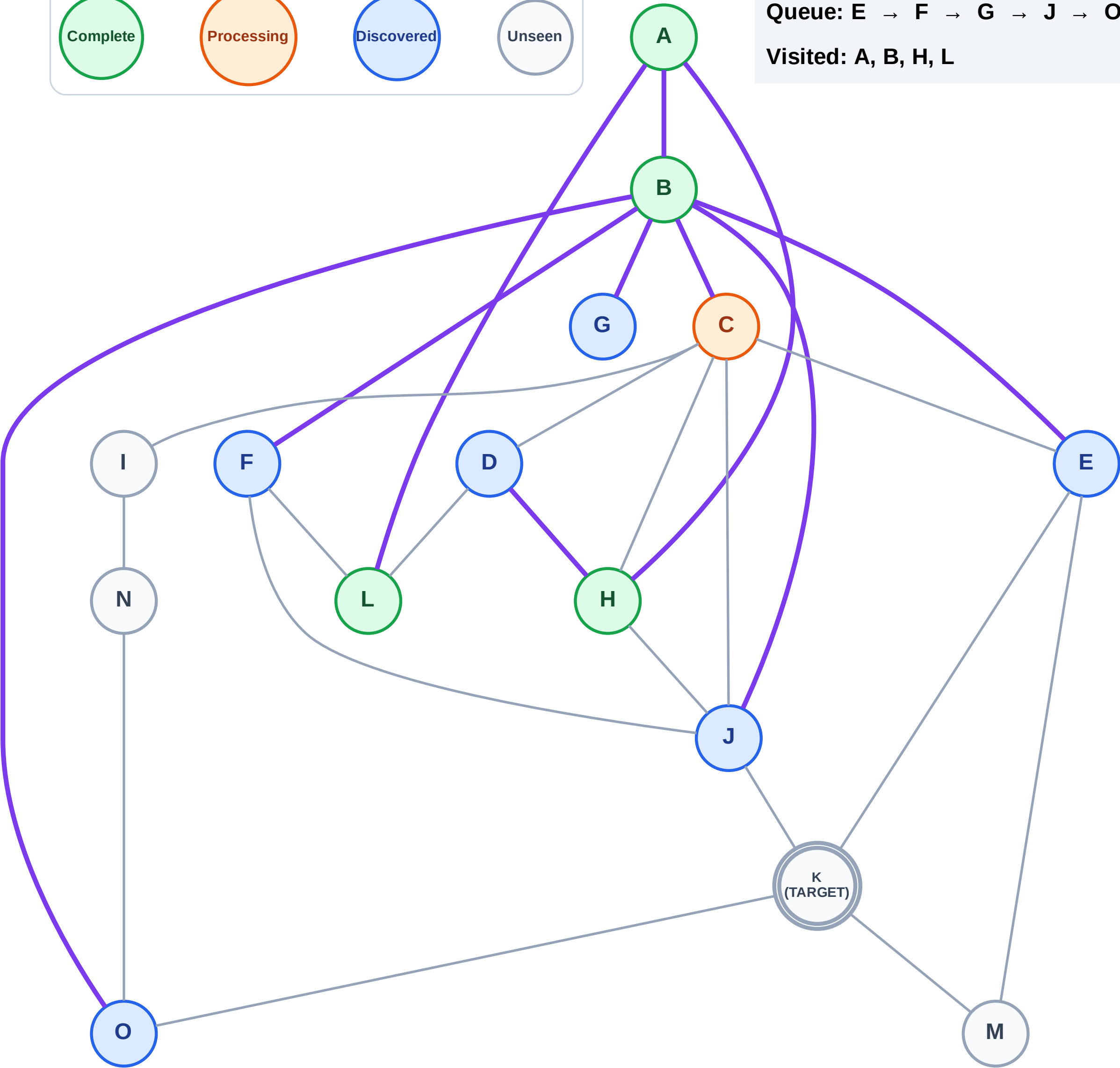
Processing

Discovered

Unseen

Queue: E → F → G → J → O → D

Visited: A, B, H, L



BFS Traversal

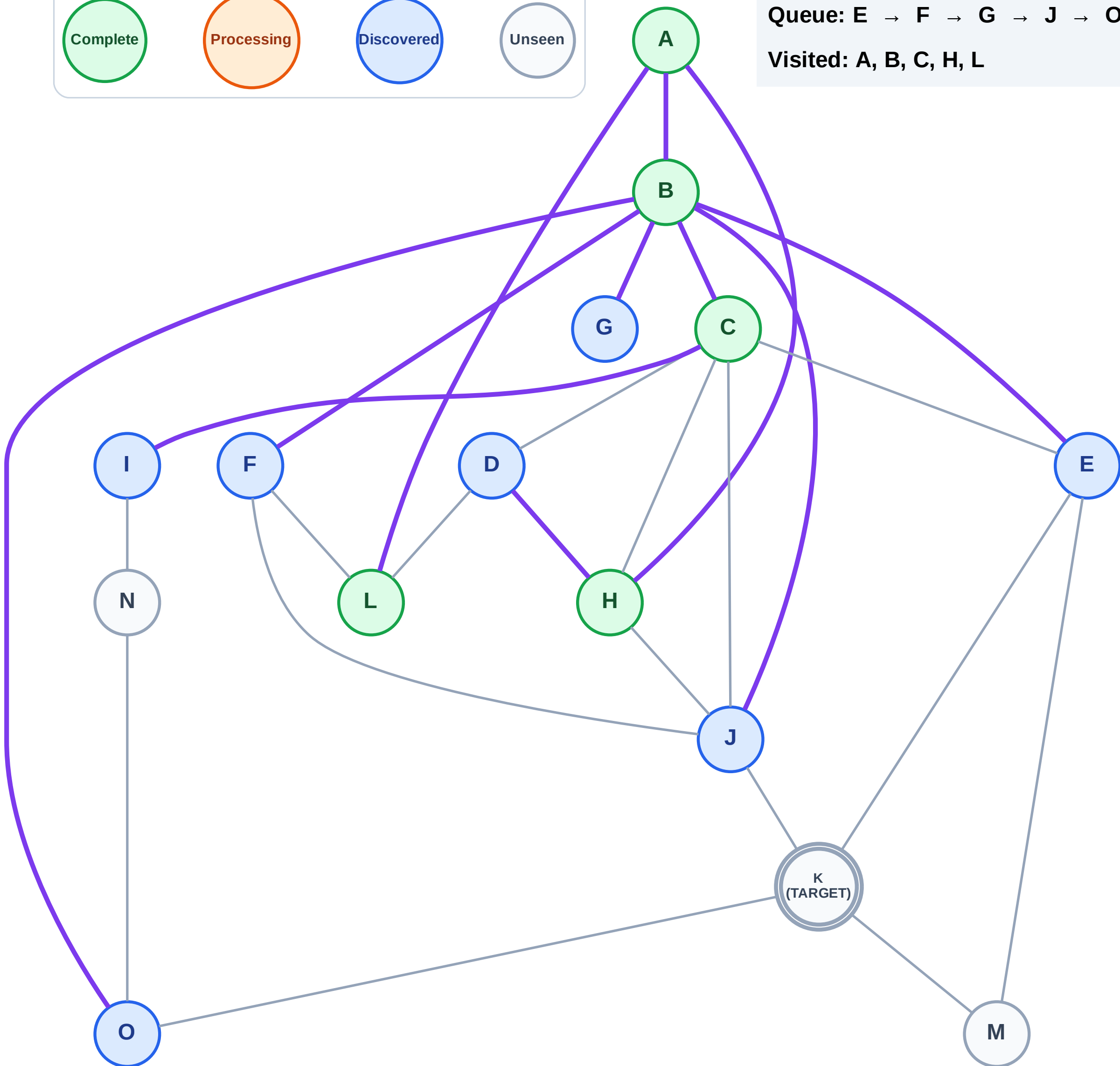
Step 11: Discover I from C

Legend



Queue: E → F → G → J → O → D → I

Visited: A, B, C, H, L



BFS Traversal

Step 12: Process node E

Legend

Complete

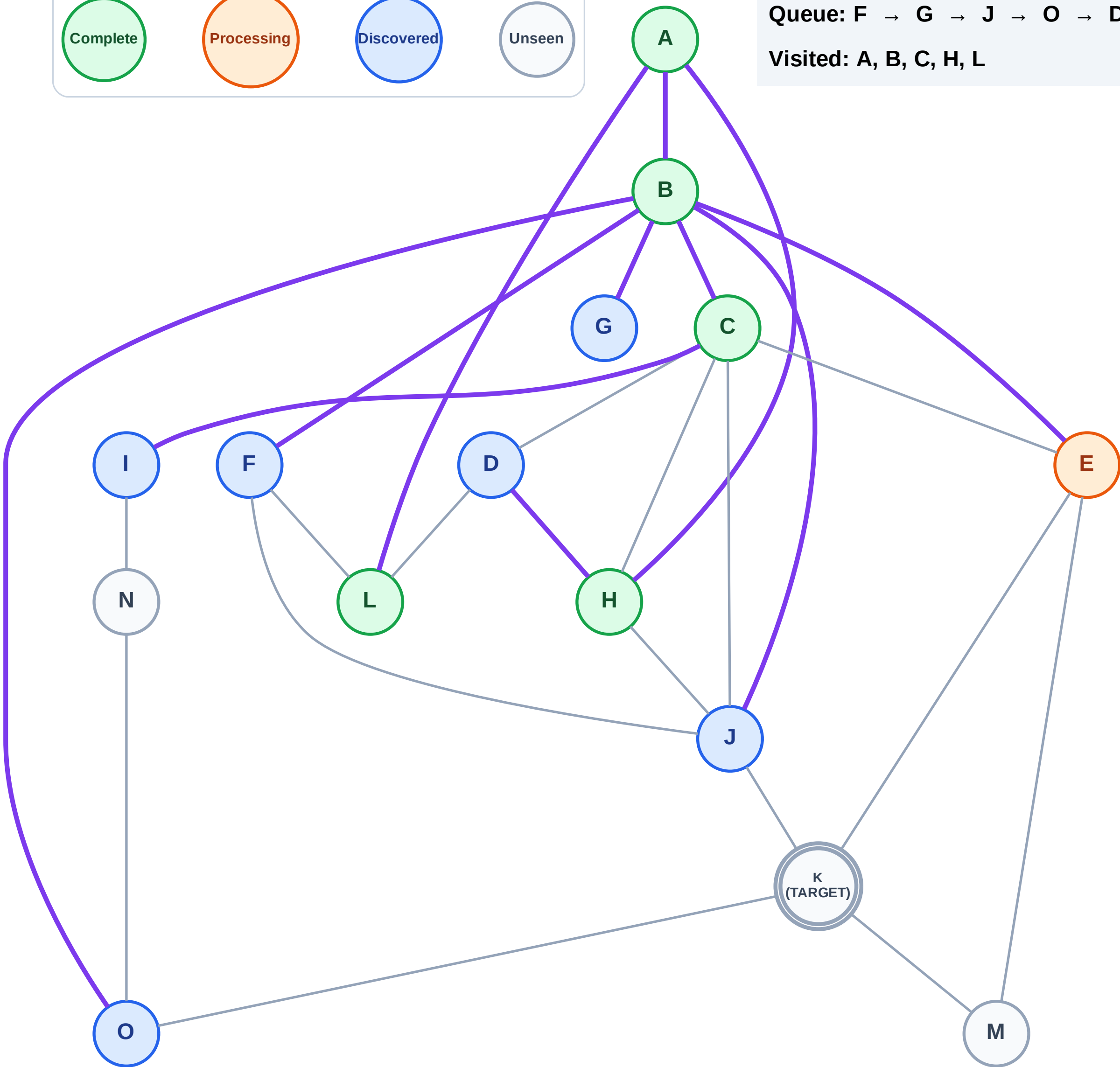
Processing

Discovered

Unseen

Queue: F → G → J → O → D → I

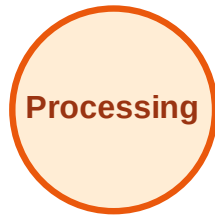
Visited: A, B, C, H, L



BFS Traversal

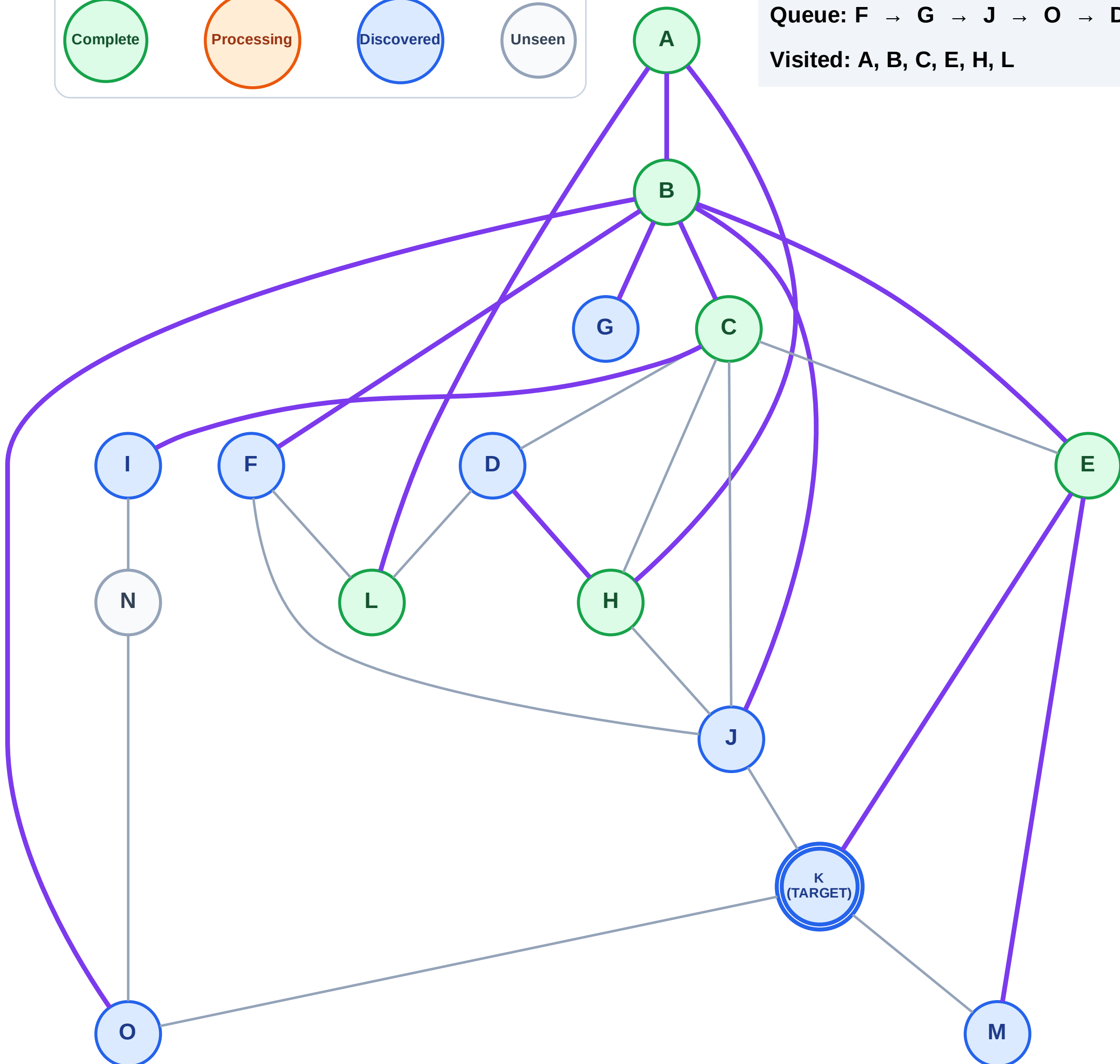
Step 13: Discover K, M from E

Legend



Queue: F → G → J → O → D → I → K → M

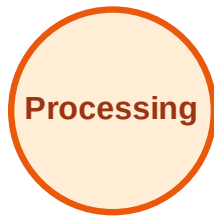
Visited: A, B, C, E, H, L



BFS Traversal

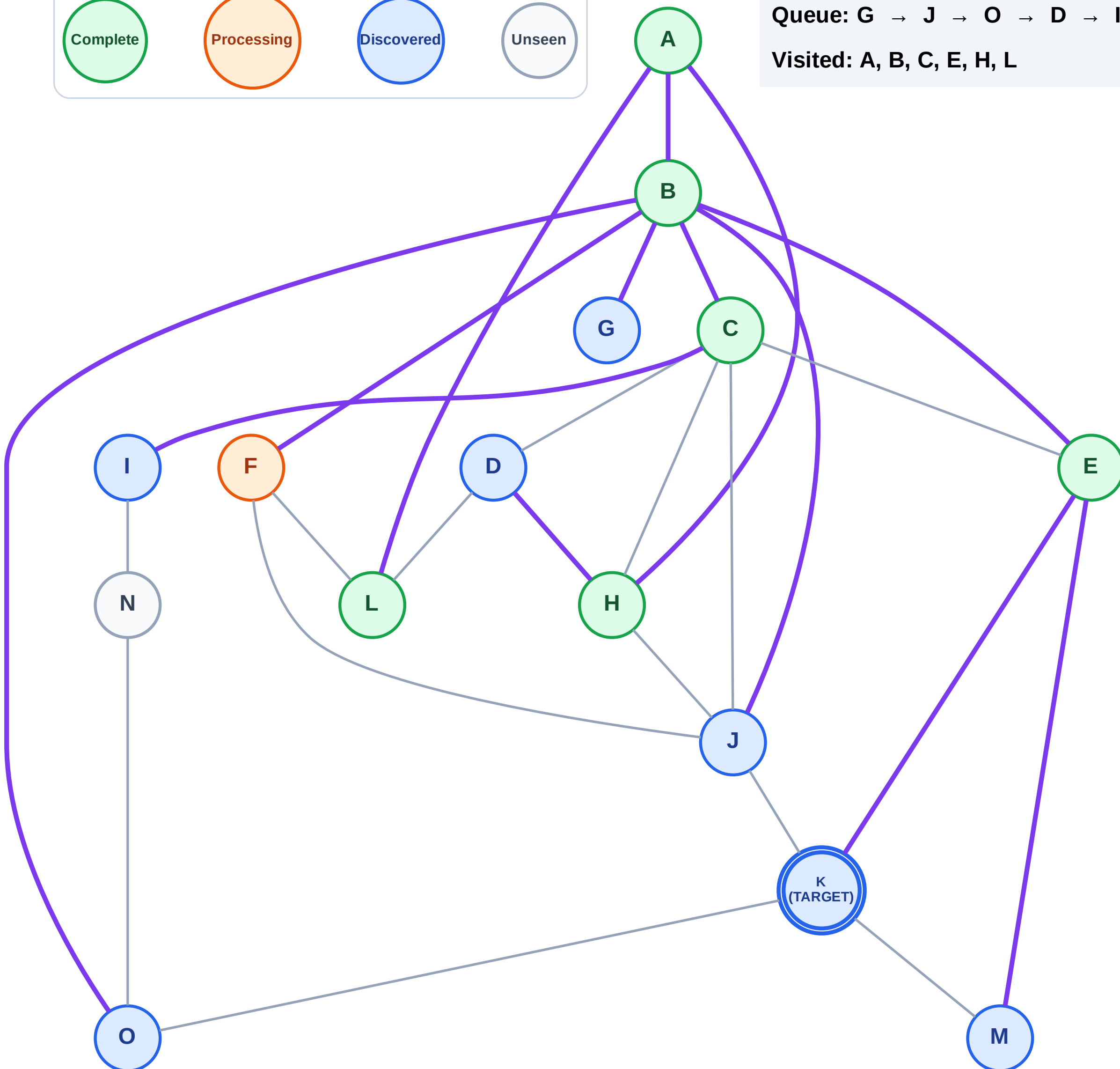
Step 14: Process node F

Legend



Queue: G → J → O → D → I → K → M

Visited: A, B, C, E, H, L



BFS Traversal

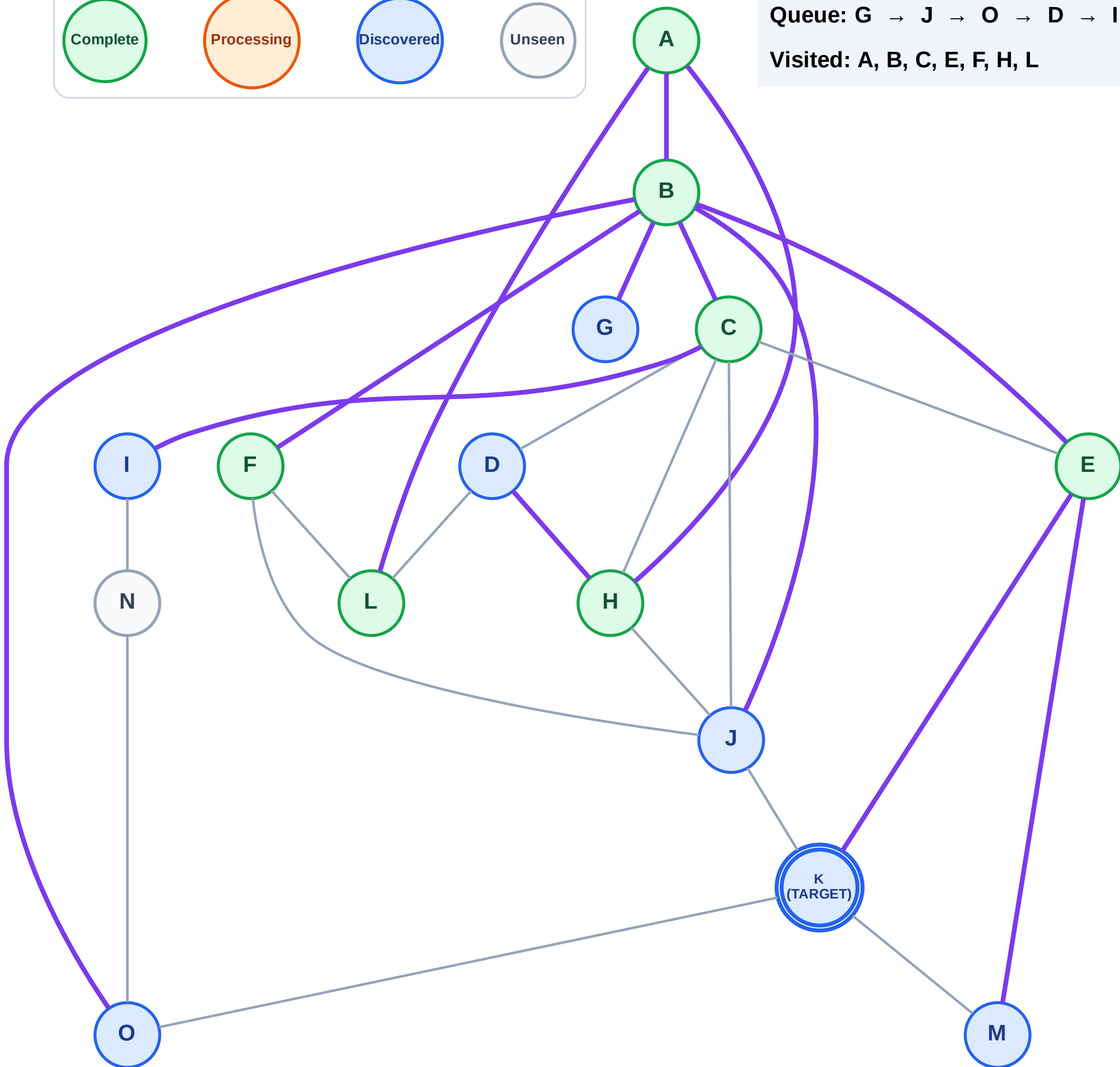
Step 15: No new neighbors from F

Legend



Queue: G → J → O → D → I → K → M

Visited: A, B, C, E, F, H, L



BFS Traversal

Step 16: Process node G

Legend

Complete

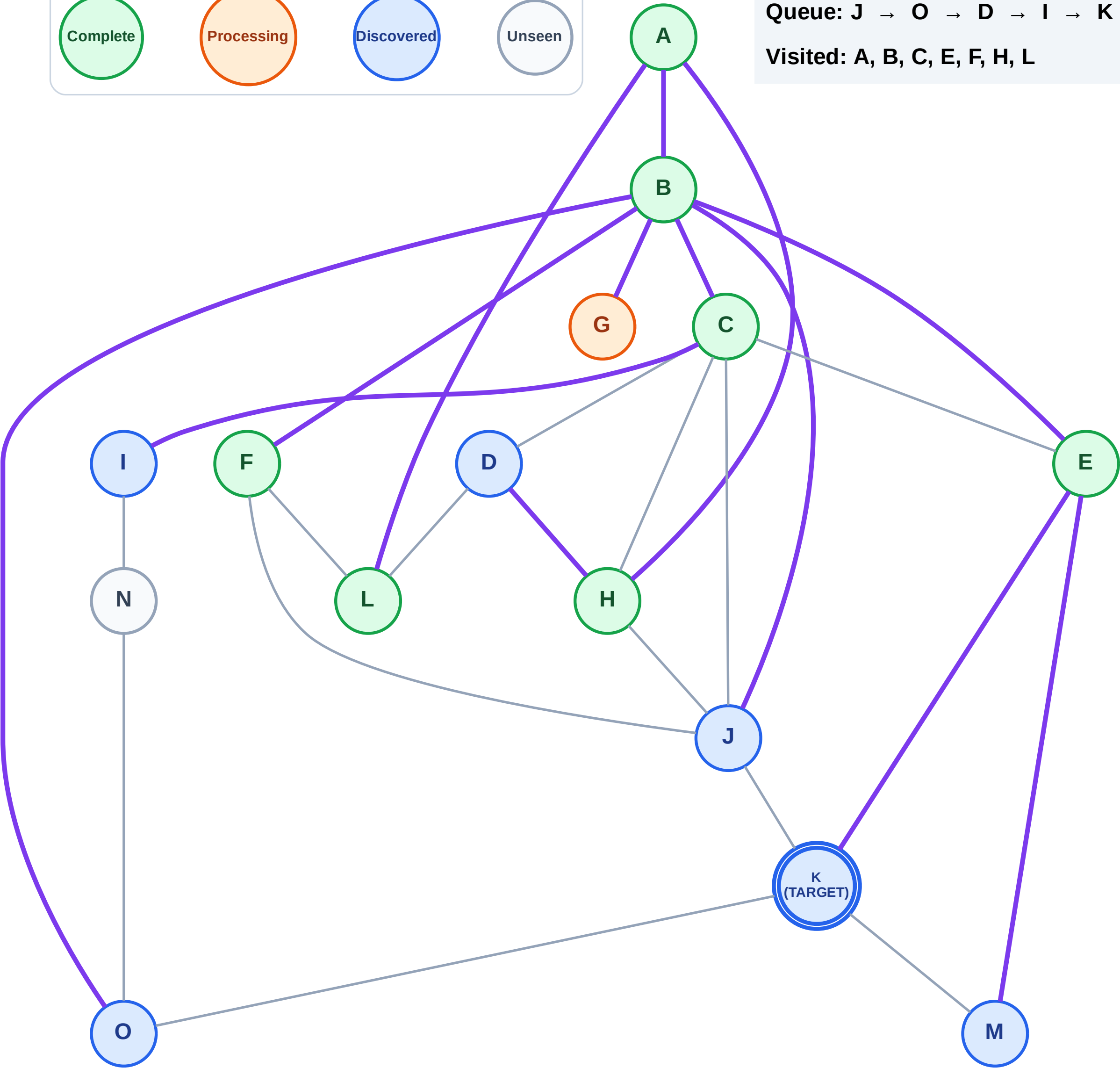
Processing

Discovered

Unseen

Queue: J → O → D → I → K → M

Visited: A, B, C, E, F, H, L



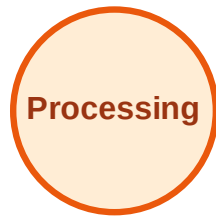
BFS Traversal

Step 17: No new neighbors from G

Legend



Complete



Processing



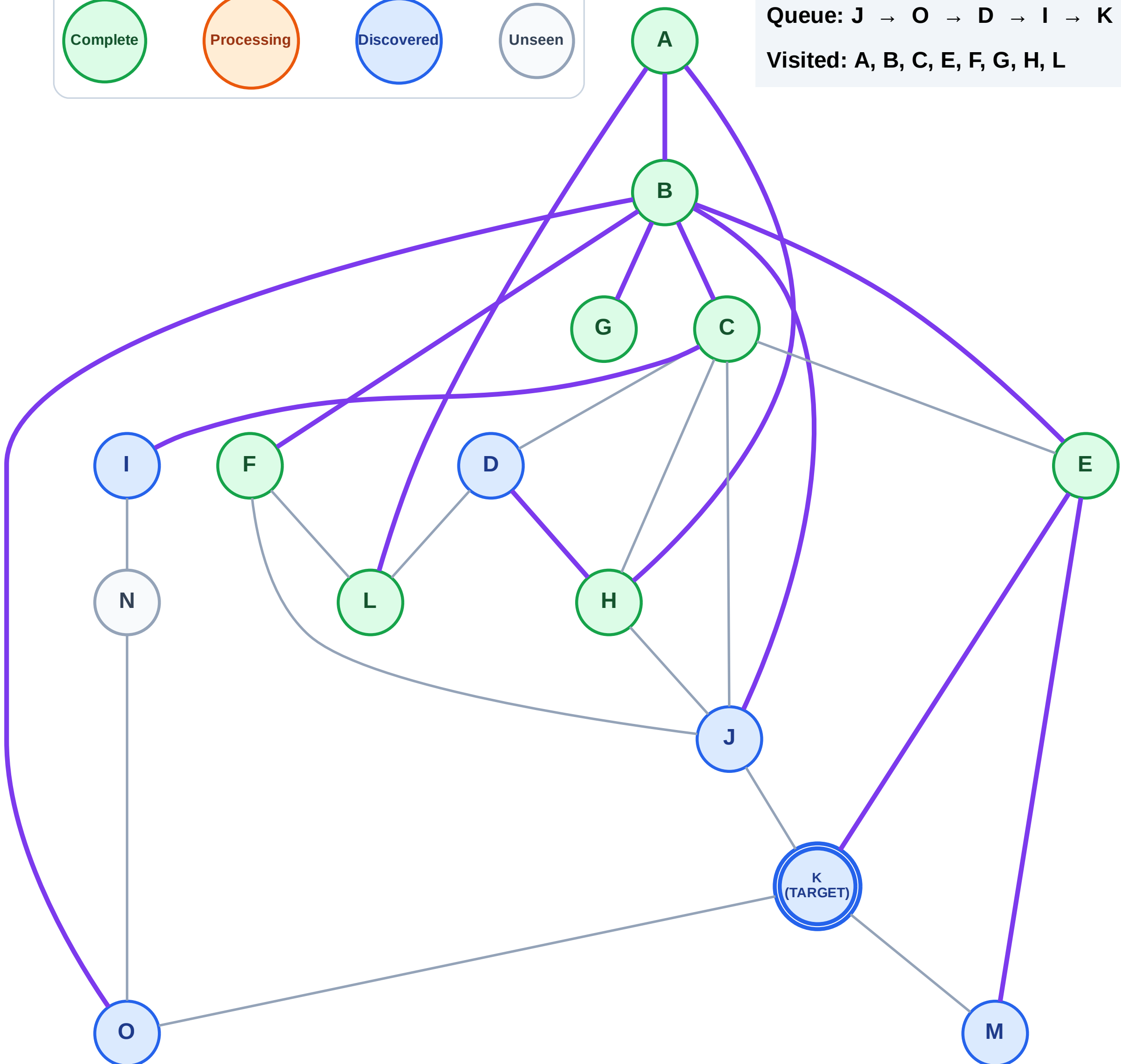
Discovered



Unseen

Queue: J → O → D → I → K → M

Visited: A, B, C, E, F, G, H, L



BFS Traversal

Step 18: Process node J

Legend

Complete

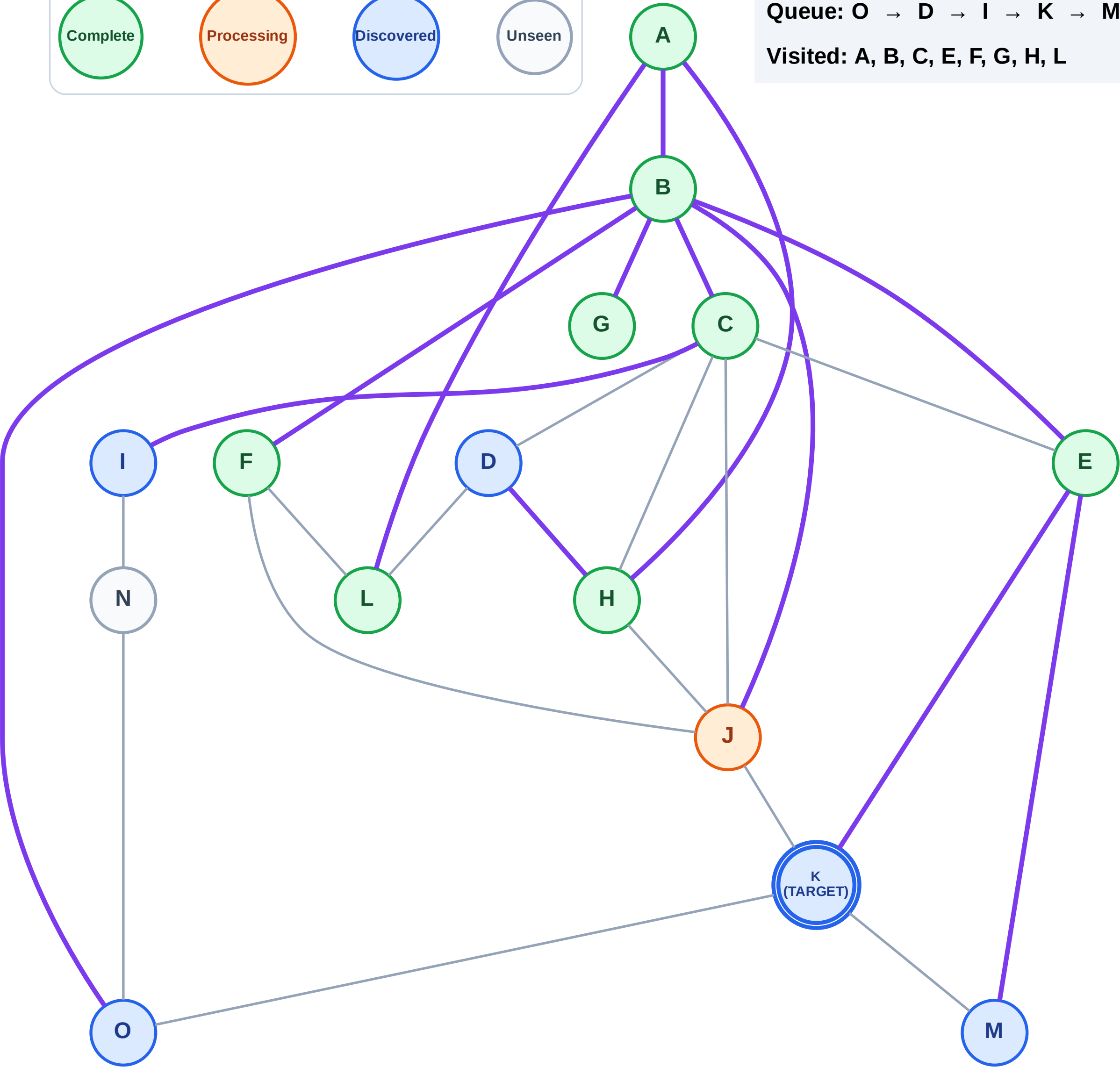
Processing

Discovered

Unseen

Queue: O → D → I → K → M

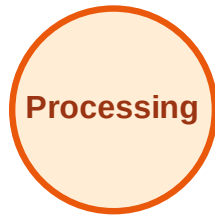
Visited: A, B, C, E, F, G, H, L



BFS Traversal

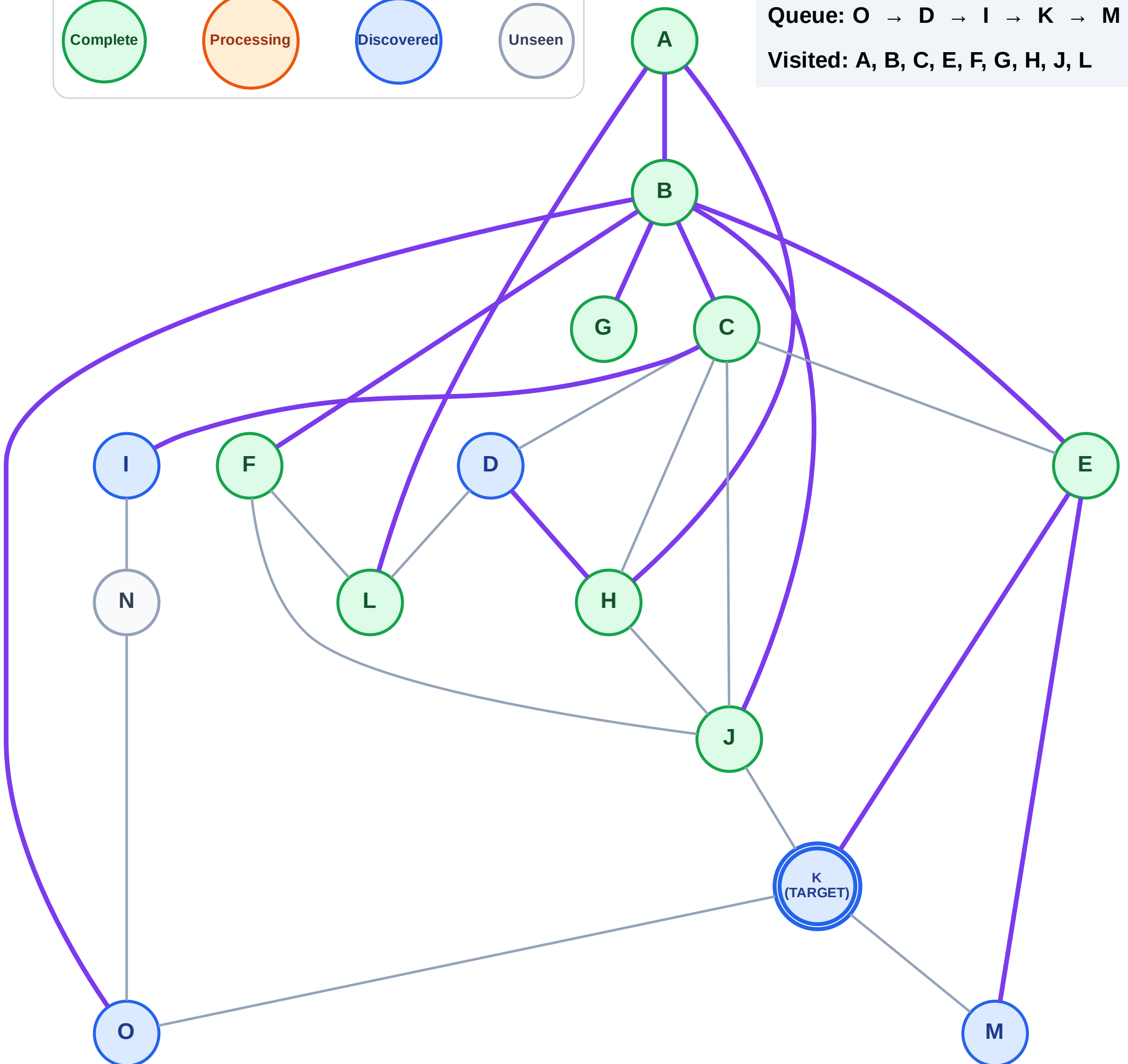
Step 19: No new neighbors from J

Legend



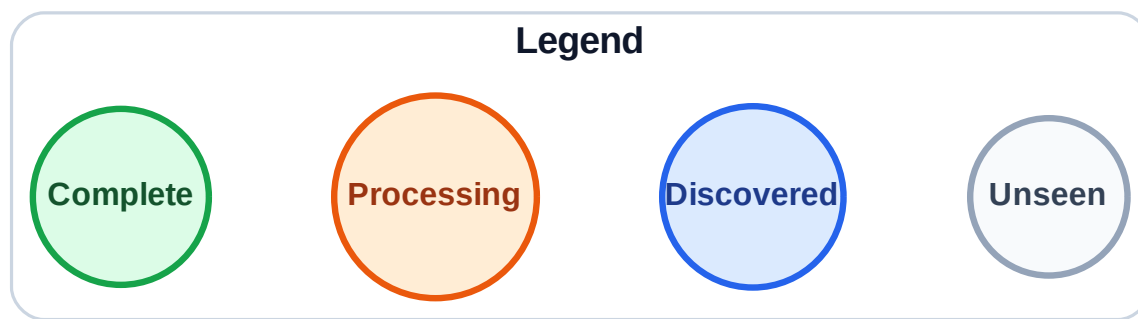
Queue: O → D → I → K → M

Visited: A, B, C, E, F, G, H, J, L



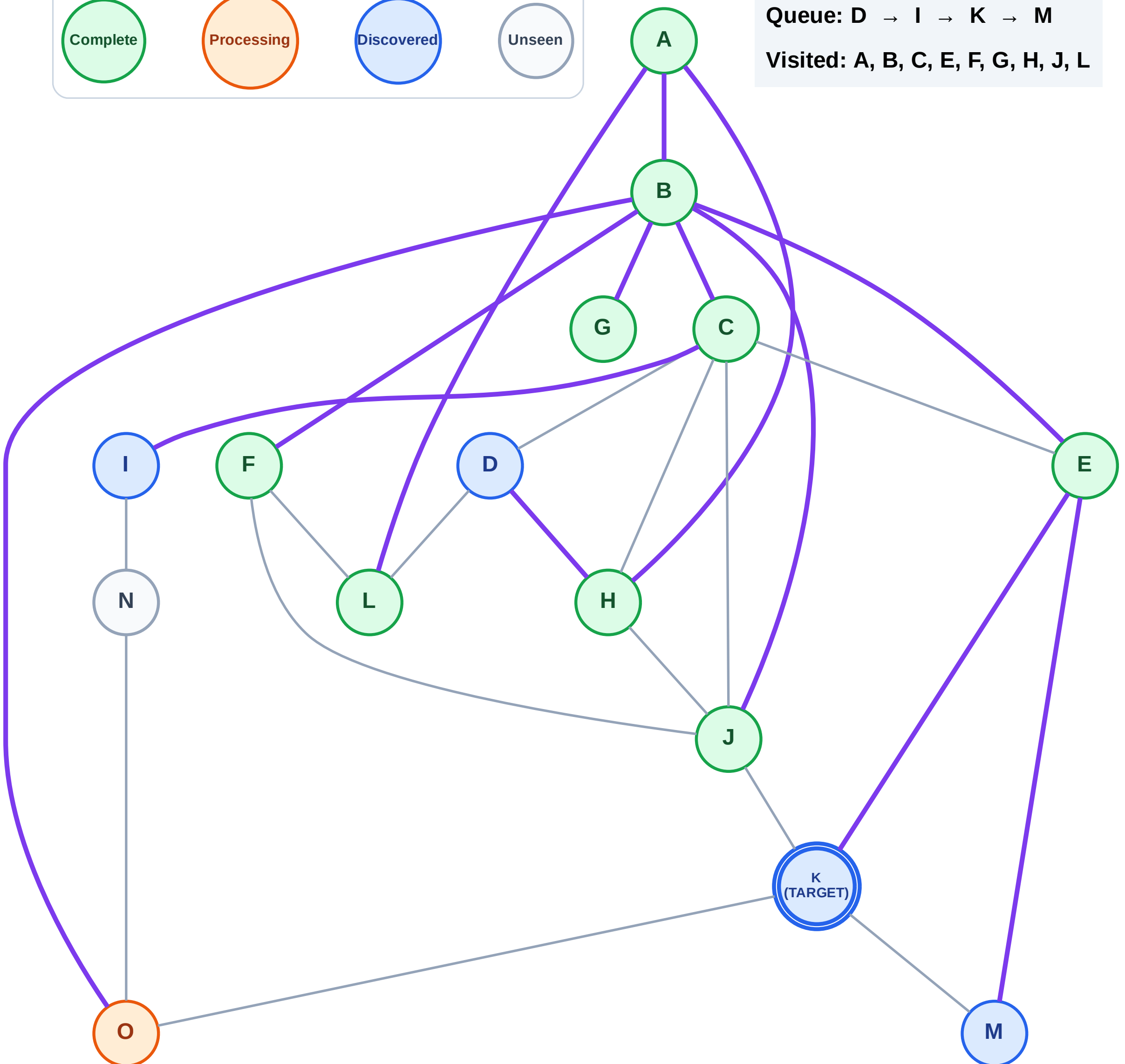
BFS Traversal

Step 20: Process node O



Queue: D → I → K → M

Visited: A, B, C, E, F, G, H, J, L



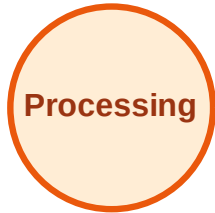
BFS Traversal

Step 21: Discover N from O

Legend



Complete



Processing



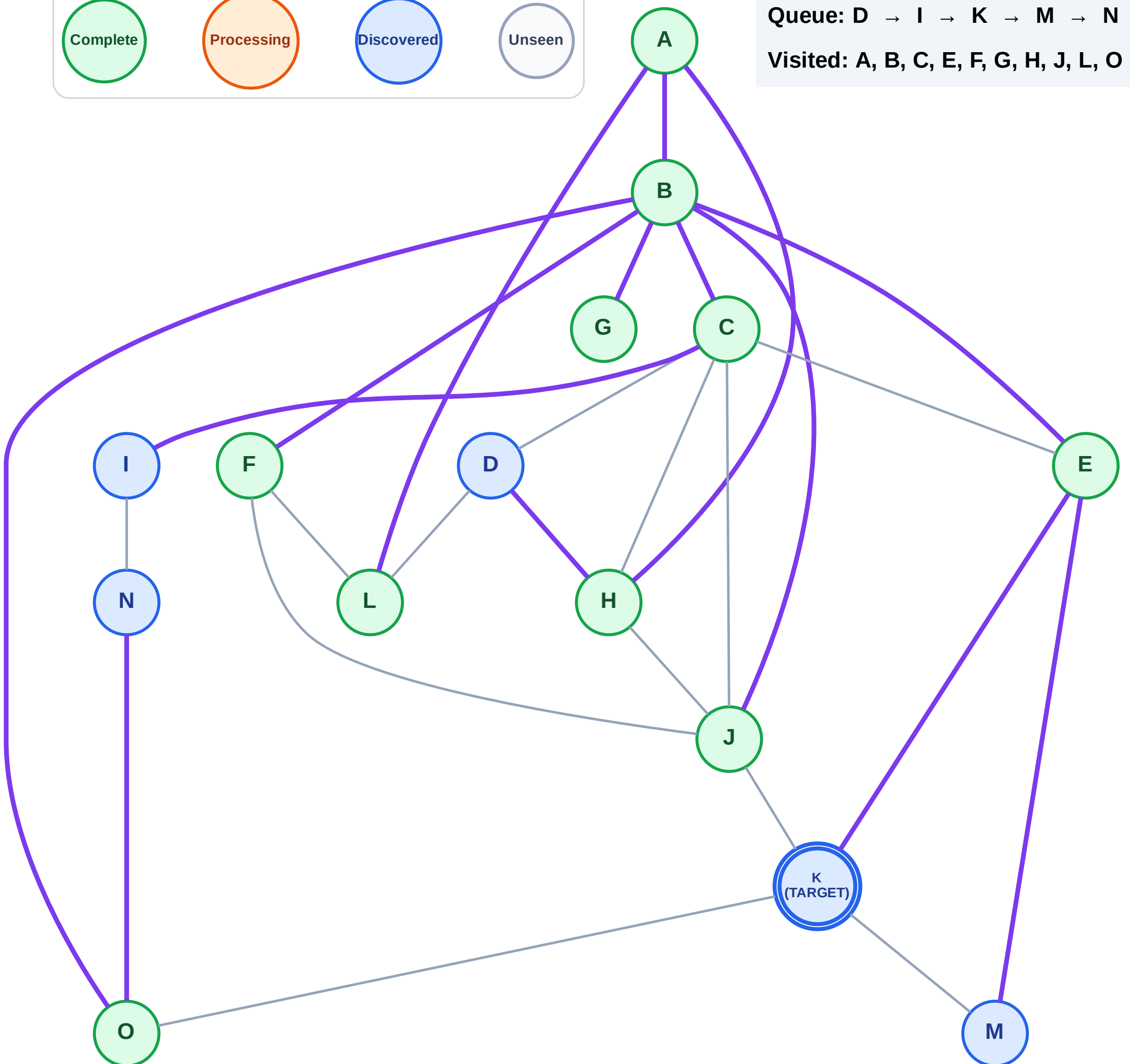
Discovered



Unseen

Queue: D → I → K → M → N

Visited: A, B, C, E, F, G, H, J, L, O



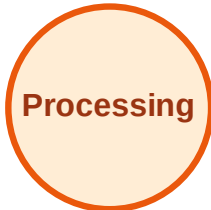
BFS Traversal

Step 22: Process node D

Legend



Complete



Processing



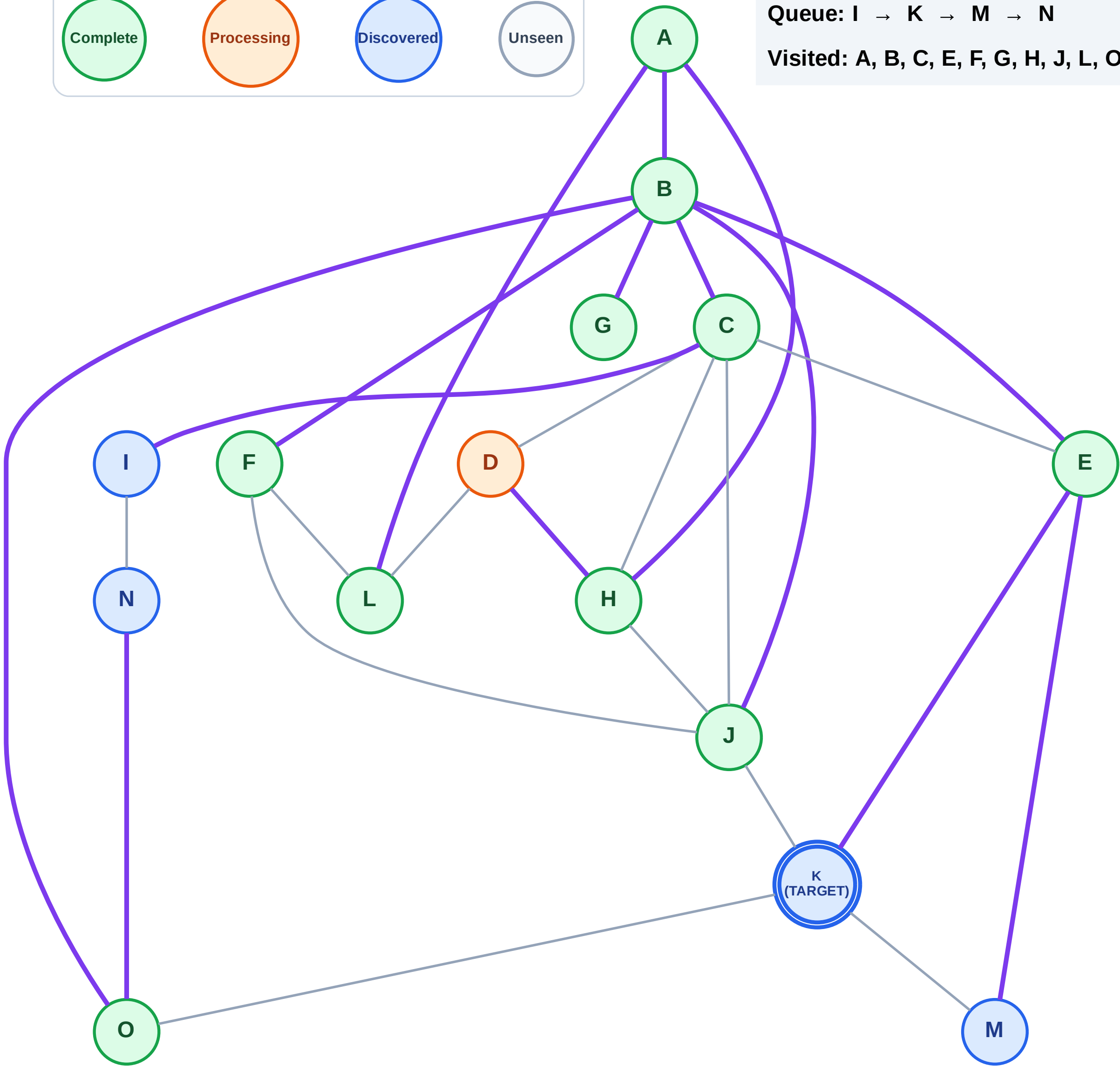
Discovered



Unseen

Queue: I → K → M → N

Visited: A, B, C, E, F, G, H, J, L, O



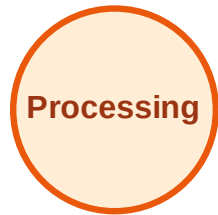
BFS Traversal

Step 23: No new neighbors from D

Legend



Complete



Processing



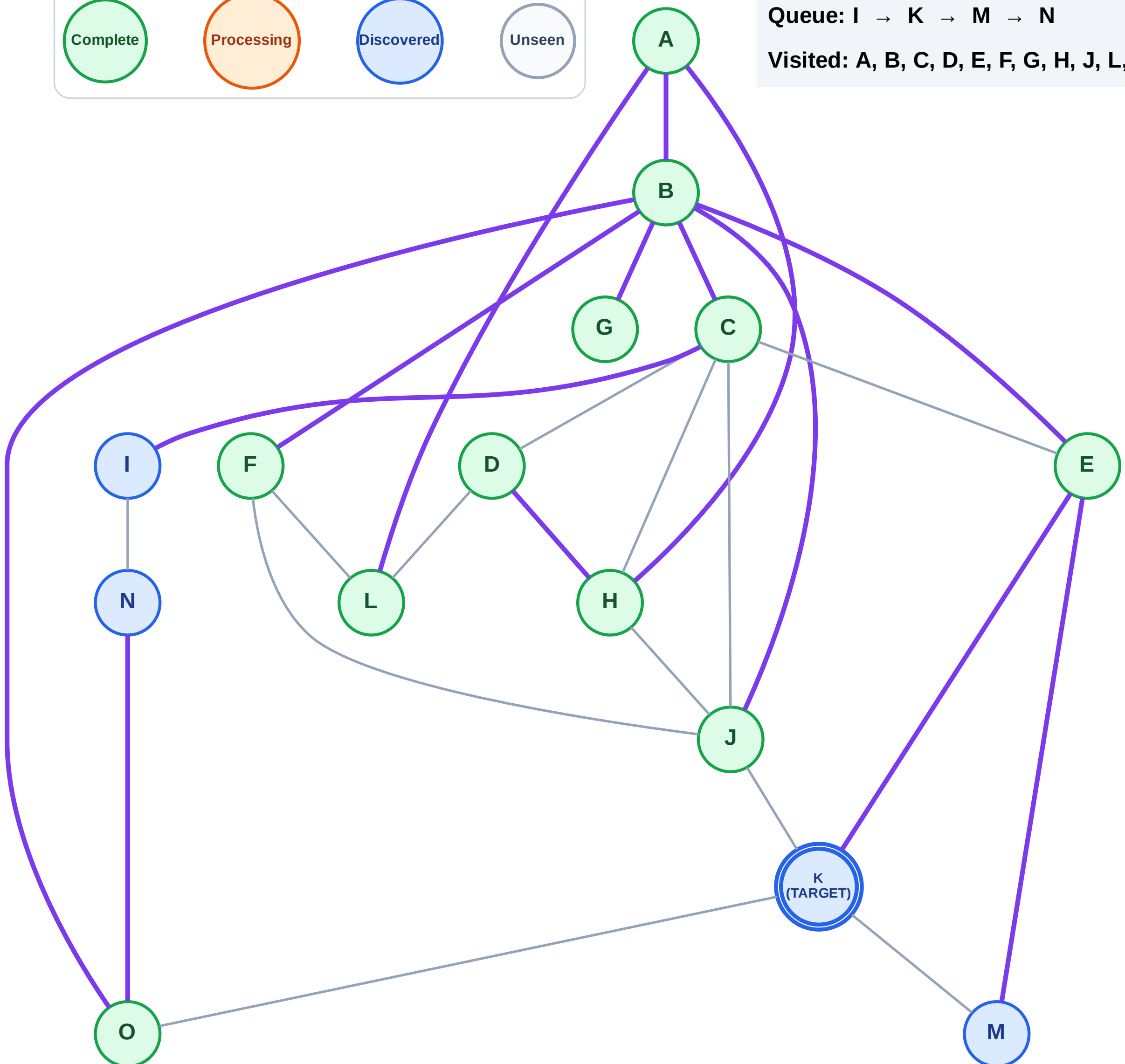
Discovered



Unseen

Queue: I → K → M → N

Visited: A, B, C, D, E, F, G, H, J, L, O



BFS Traversal

Step 24: Process node I

Legend

Complete

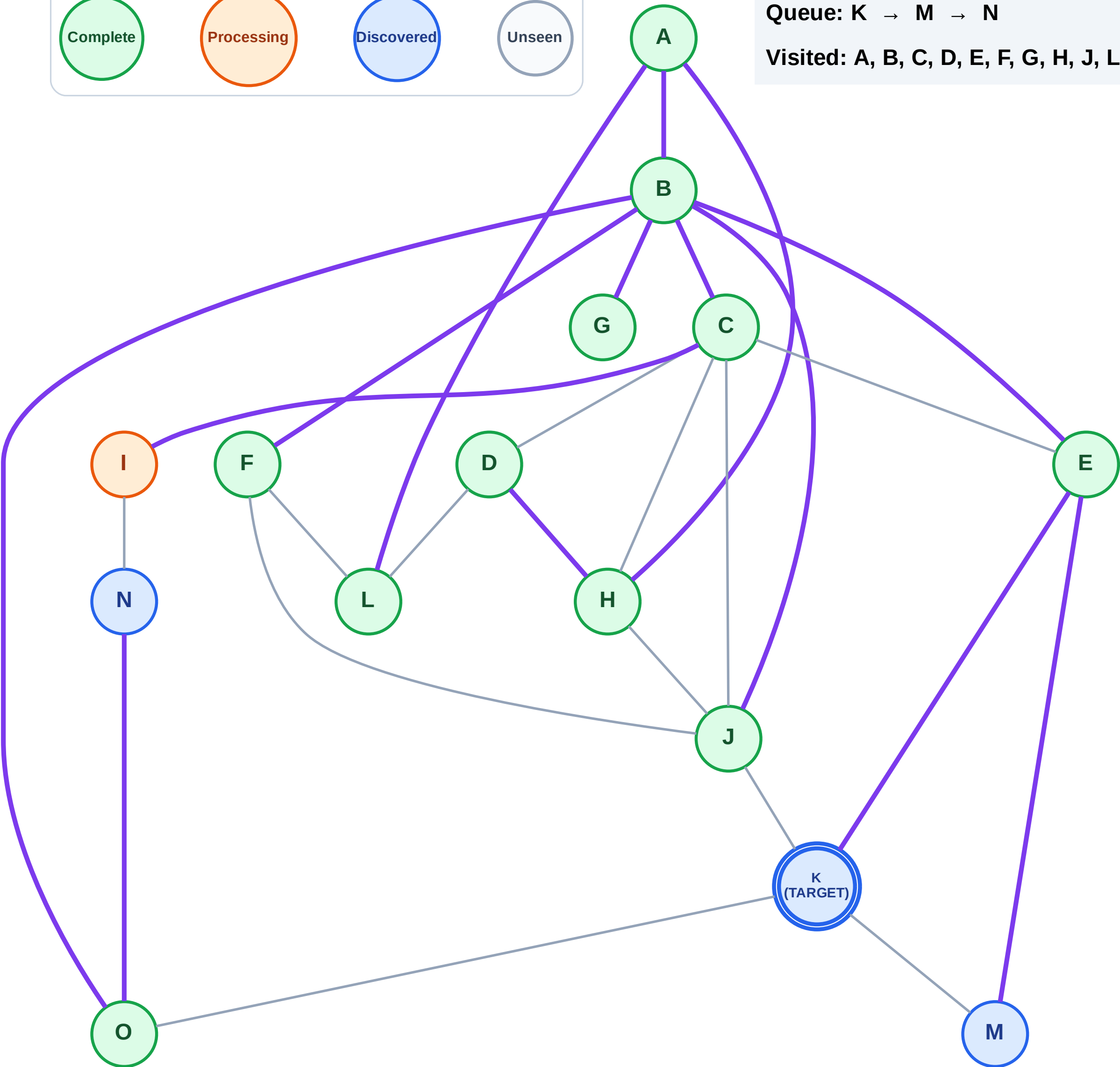
Processing

Discovered

Unseen

Queue: K → M → N

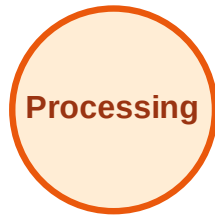
Visited: A, B, C, D, E, F, G, H, J, L, O



BFS Traversal

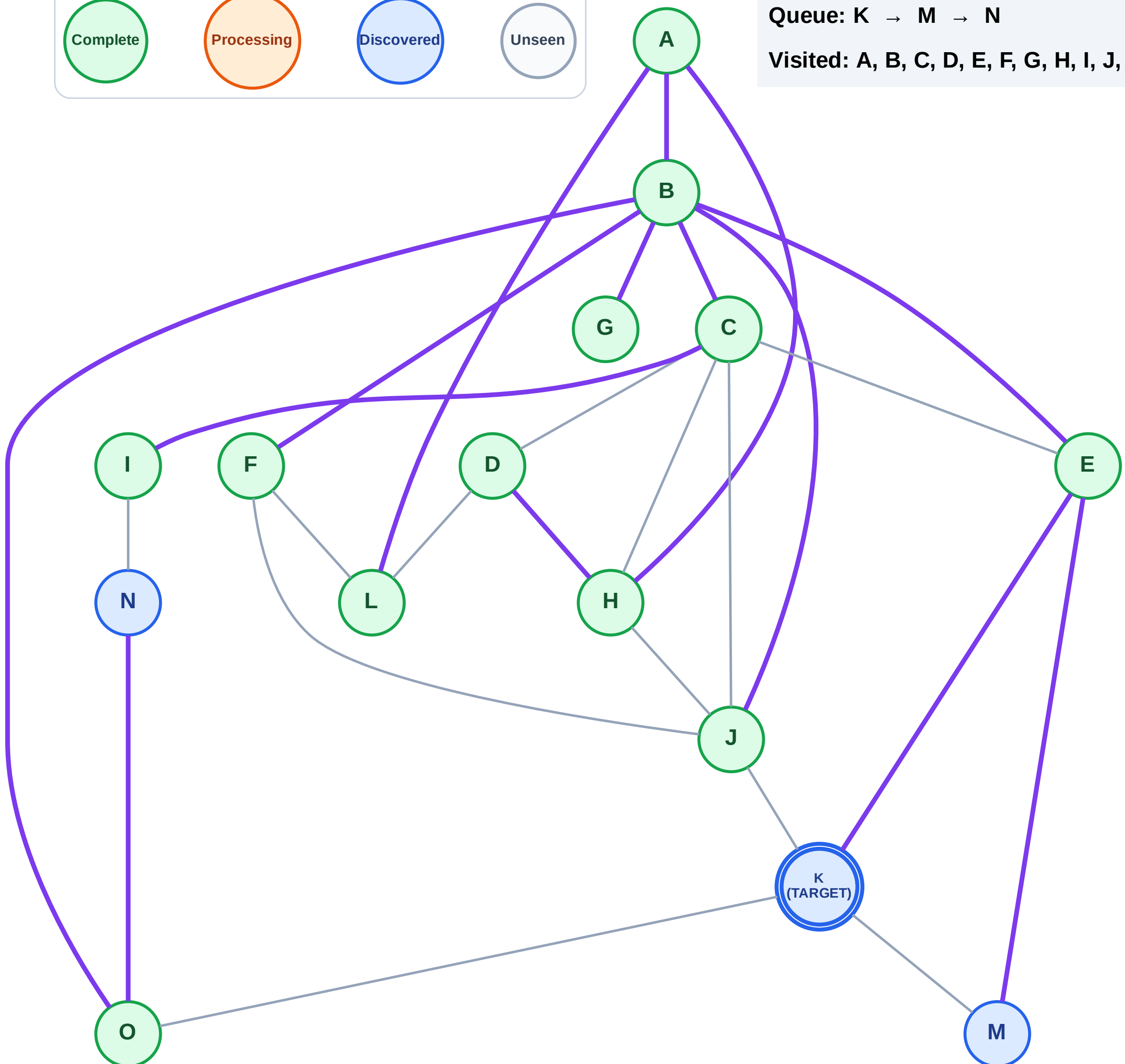
Step 25: No new neighbors from I

Legend



Queue: K → M → N

Visited: A, B, C, D, E, F, G, H, I, J, L, O



BFS Traversal

Step 26: Process node K

Legend

Complete

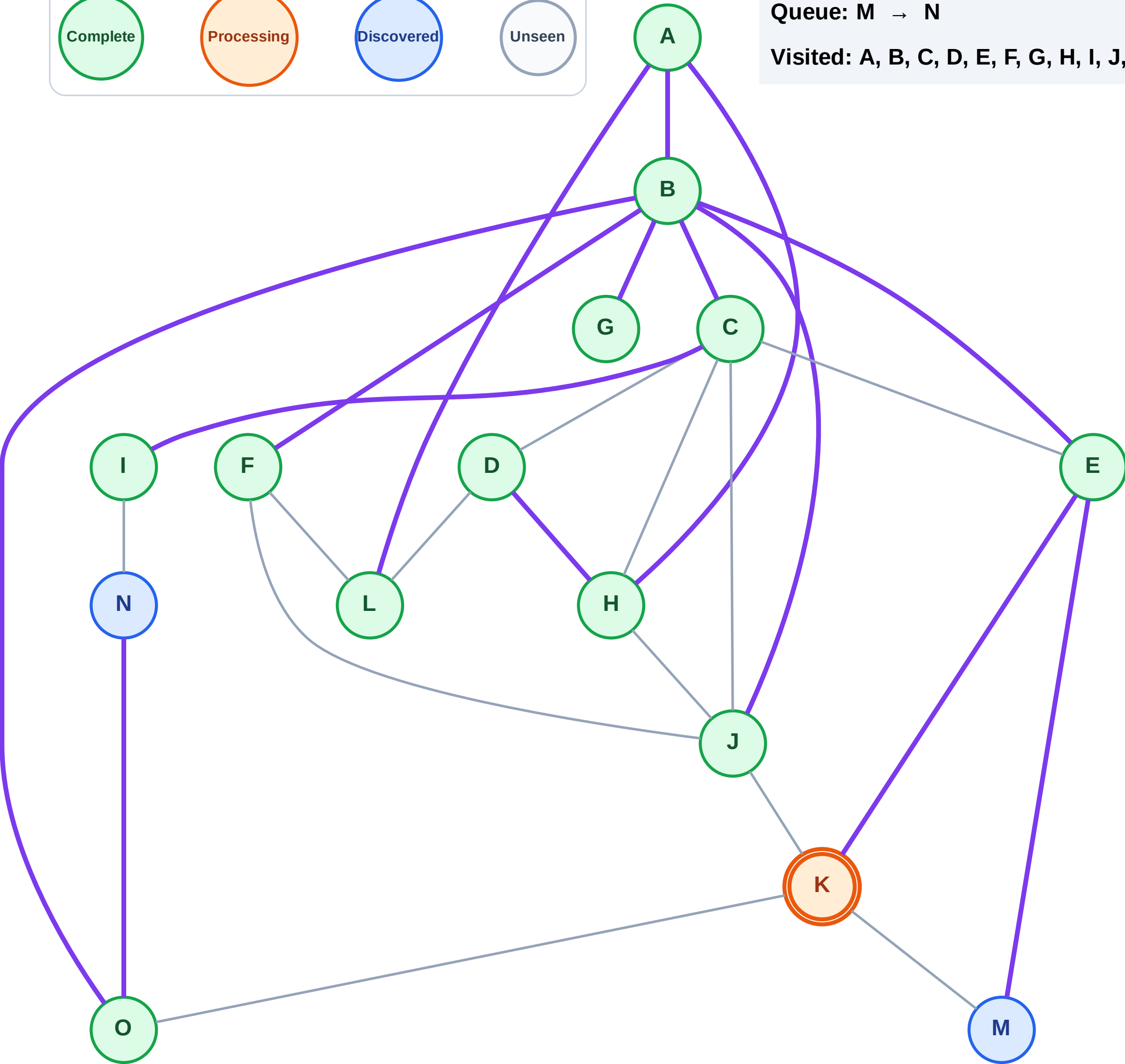
Processing

Discovered

Unseen

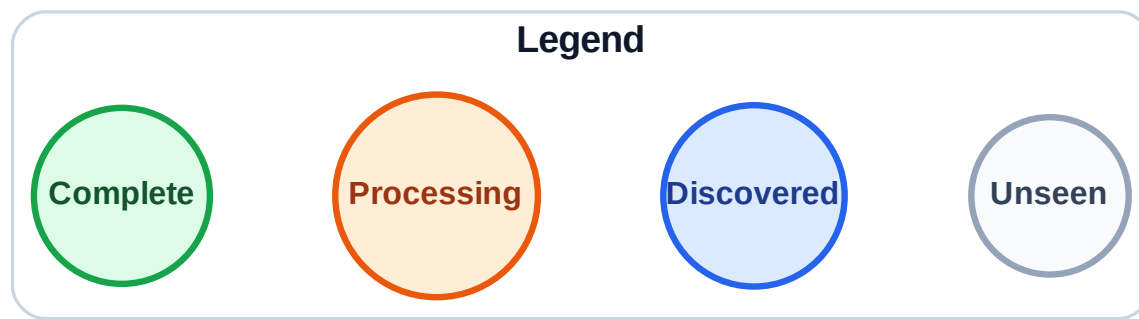
Queue: M → N

Visited: A, B, C, D, E, F, G, H, I, J, L, O



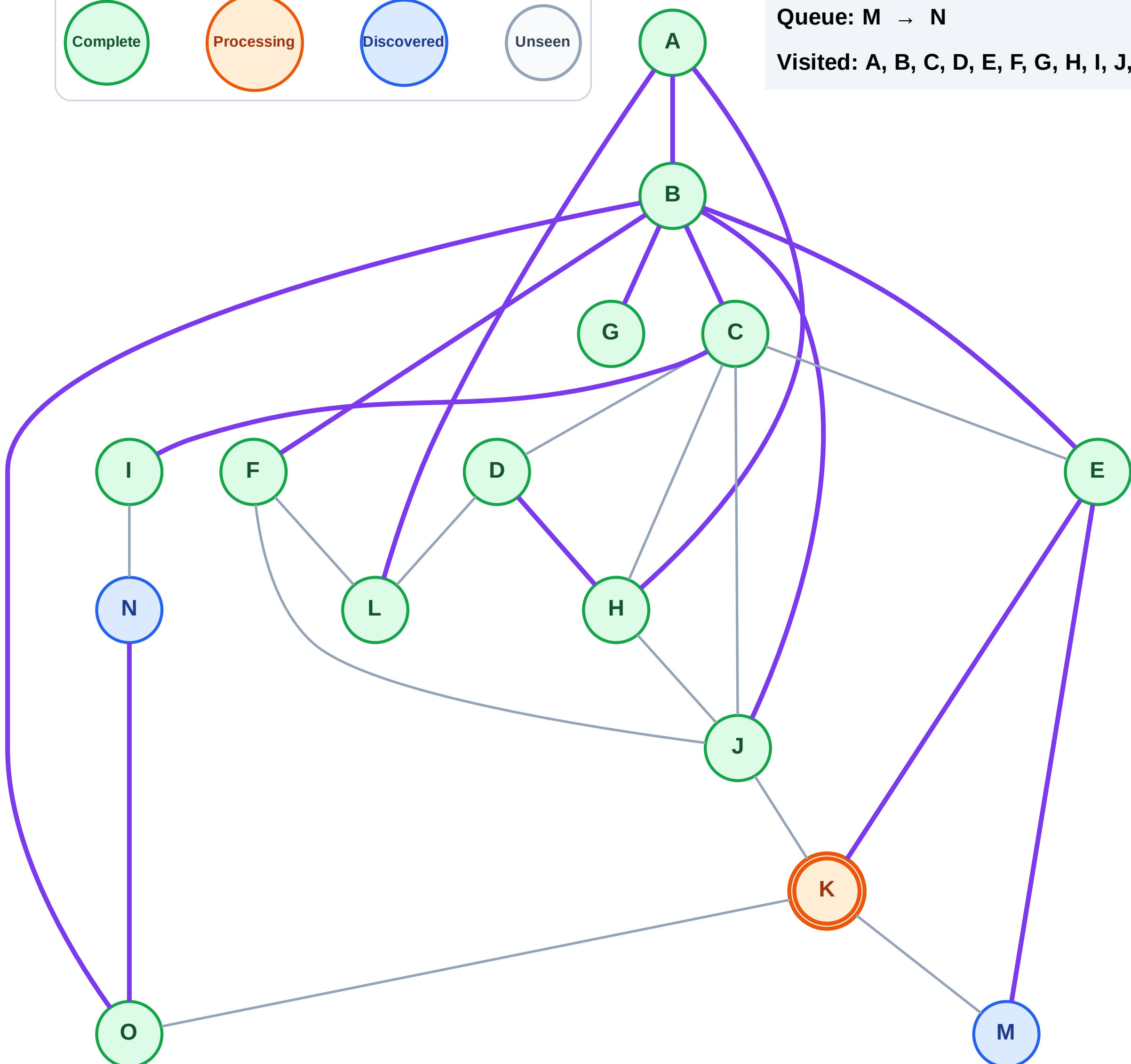
BFS Traversal

Step 27: TARGET NODE K REACHED! Path (3 hops): A → B → E → K



Queue: $M \rightarrow N$

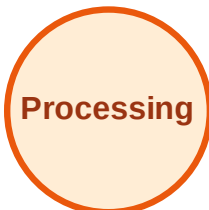
Visited: A, B, C, D, E, F, G, H, I, J, K, L, O



DFS Traversal

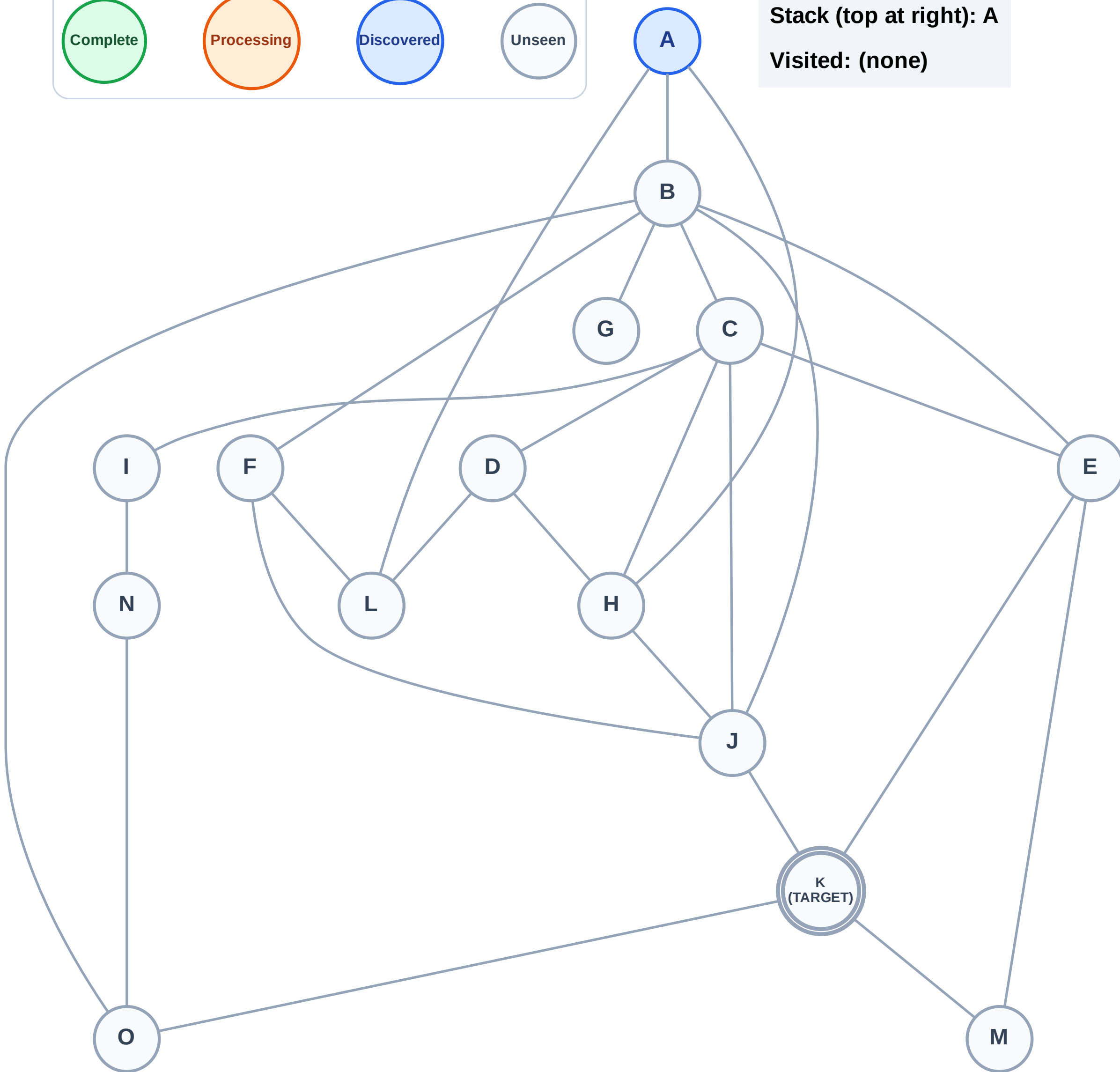
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

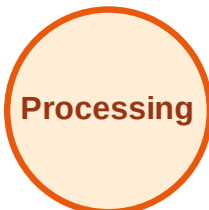
Visited: (none)



DFS Traversal

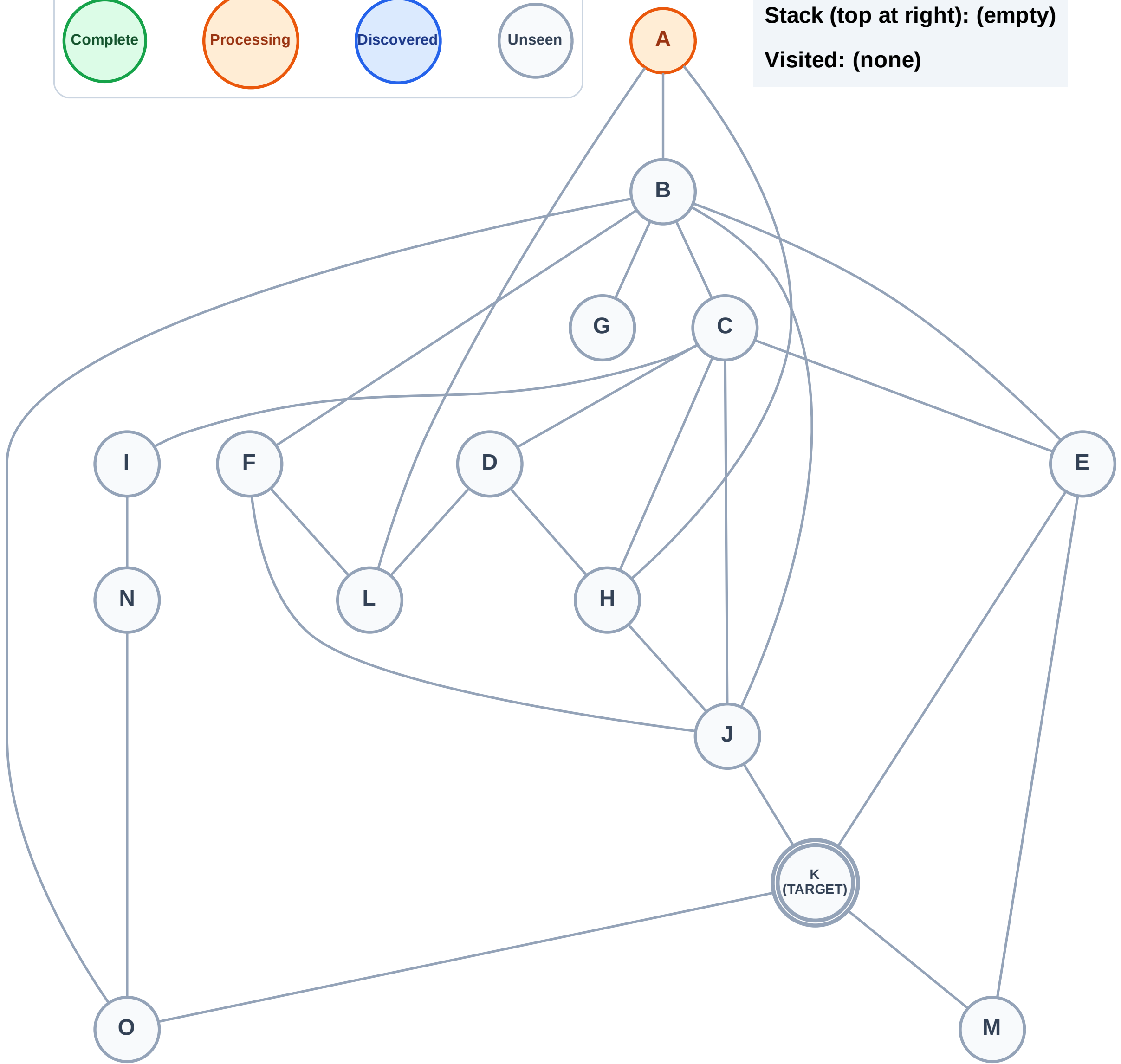
Step 2: Process node A

Legend



Stack (top at right): (empty)

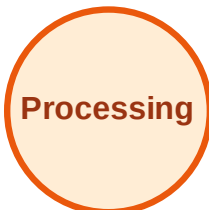
Visited: (none)



DFS Traversal

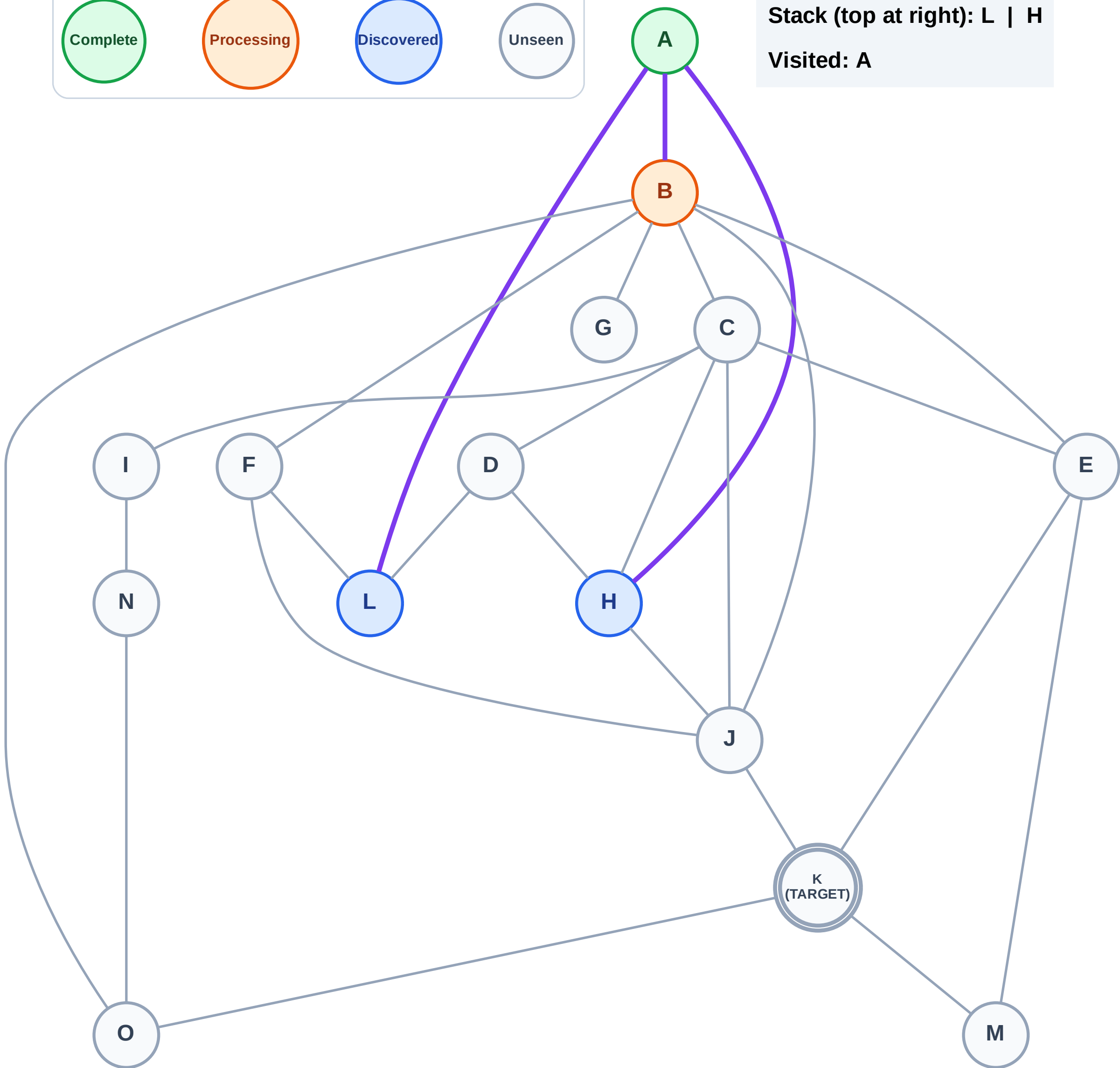
Step 4: Process node B

Legend



Stack (top at right): L | H

Visited: A



DFS Traversal

Step 5: Discover O, J, G, F, E, C from B

Legend

Complete

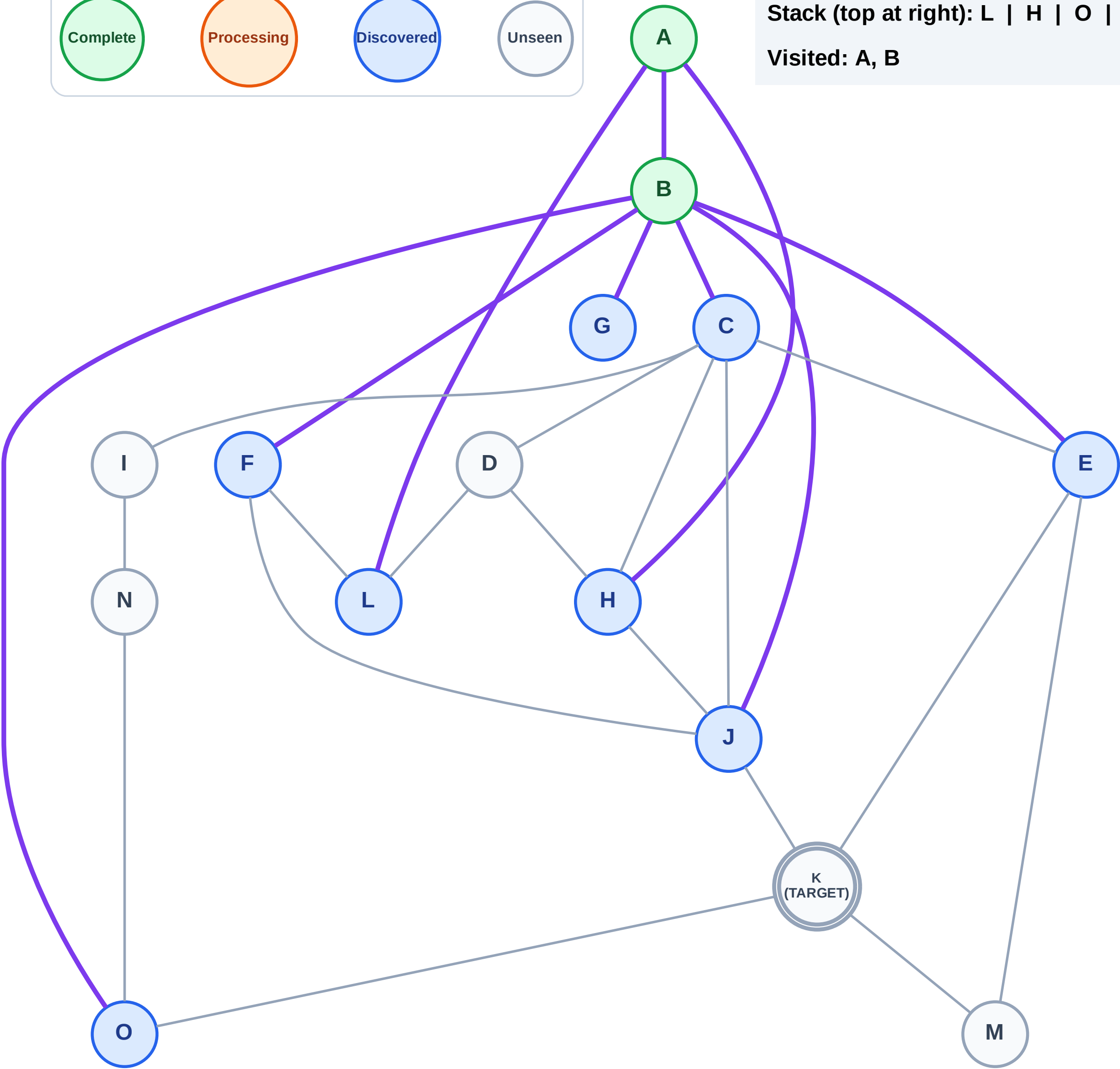
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



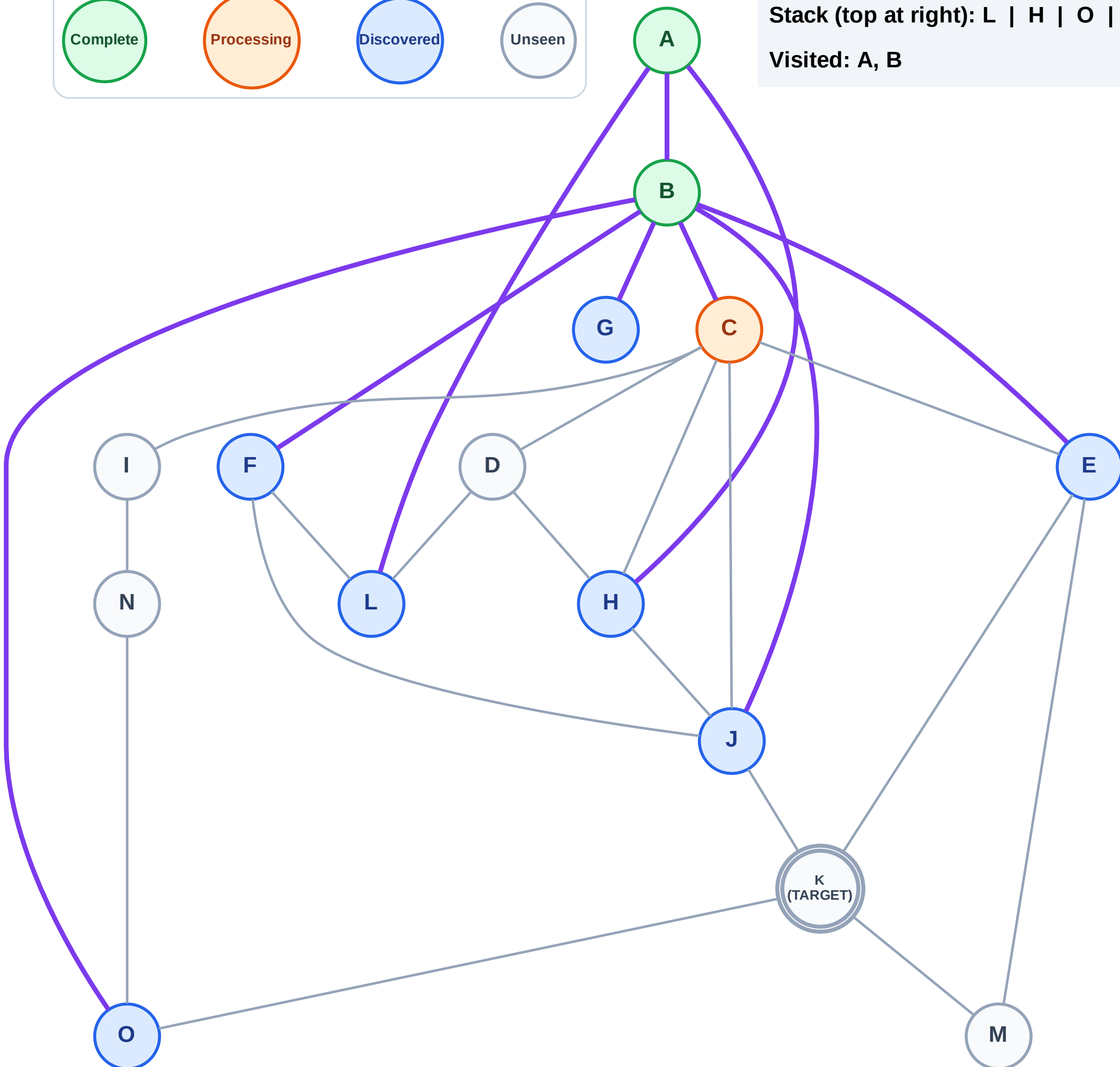
Discovered



Unseen

Stack (top at right): L | H | O | J | G | F | E

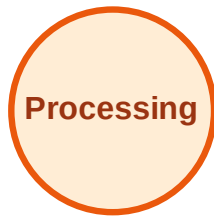
Visited: A, B



DFS Traversal

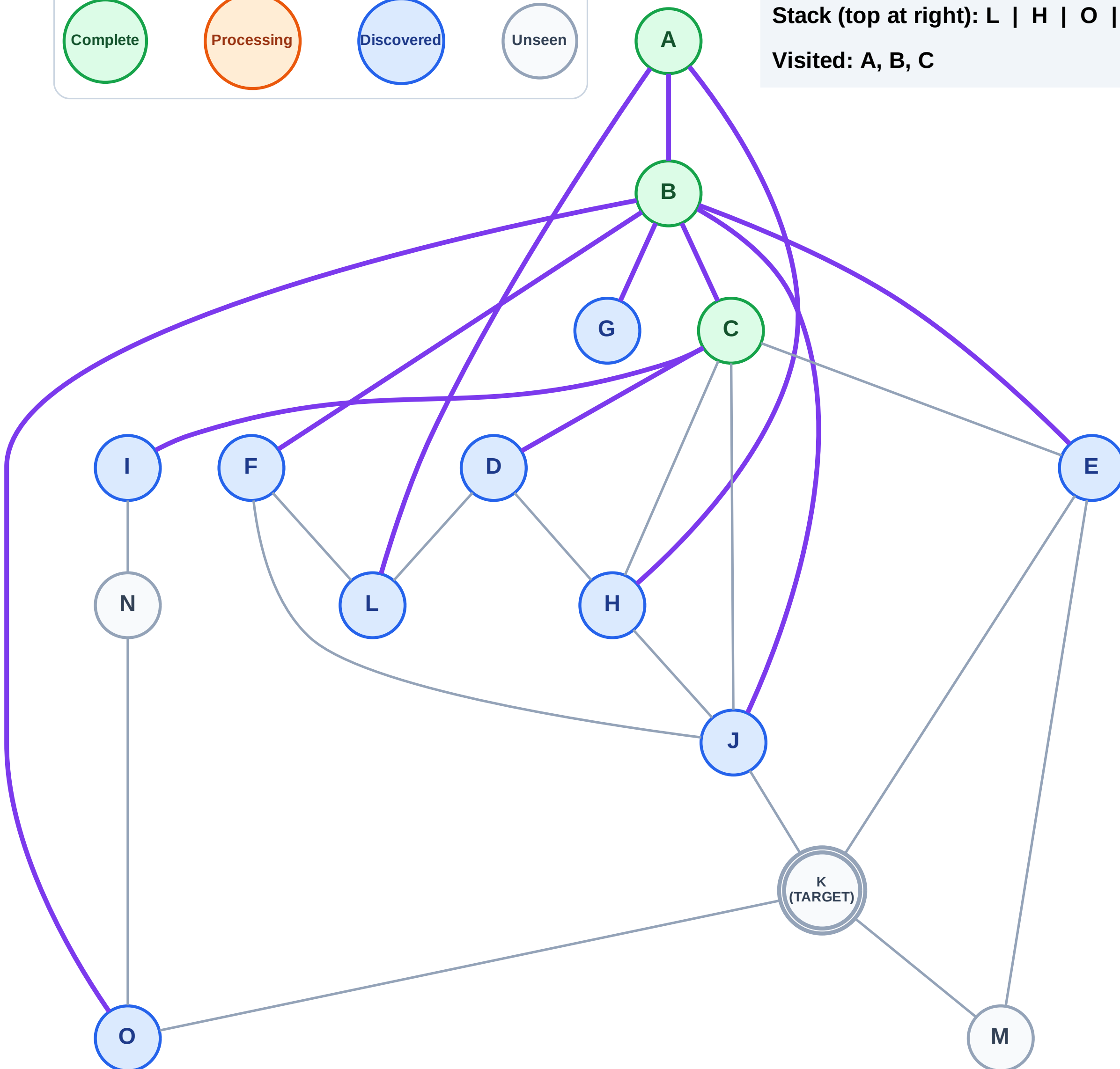
Step 7: Discover I, D from C

Legend



Stack (top at right): L | H | O | J | G | F | E | I | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

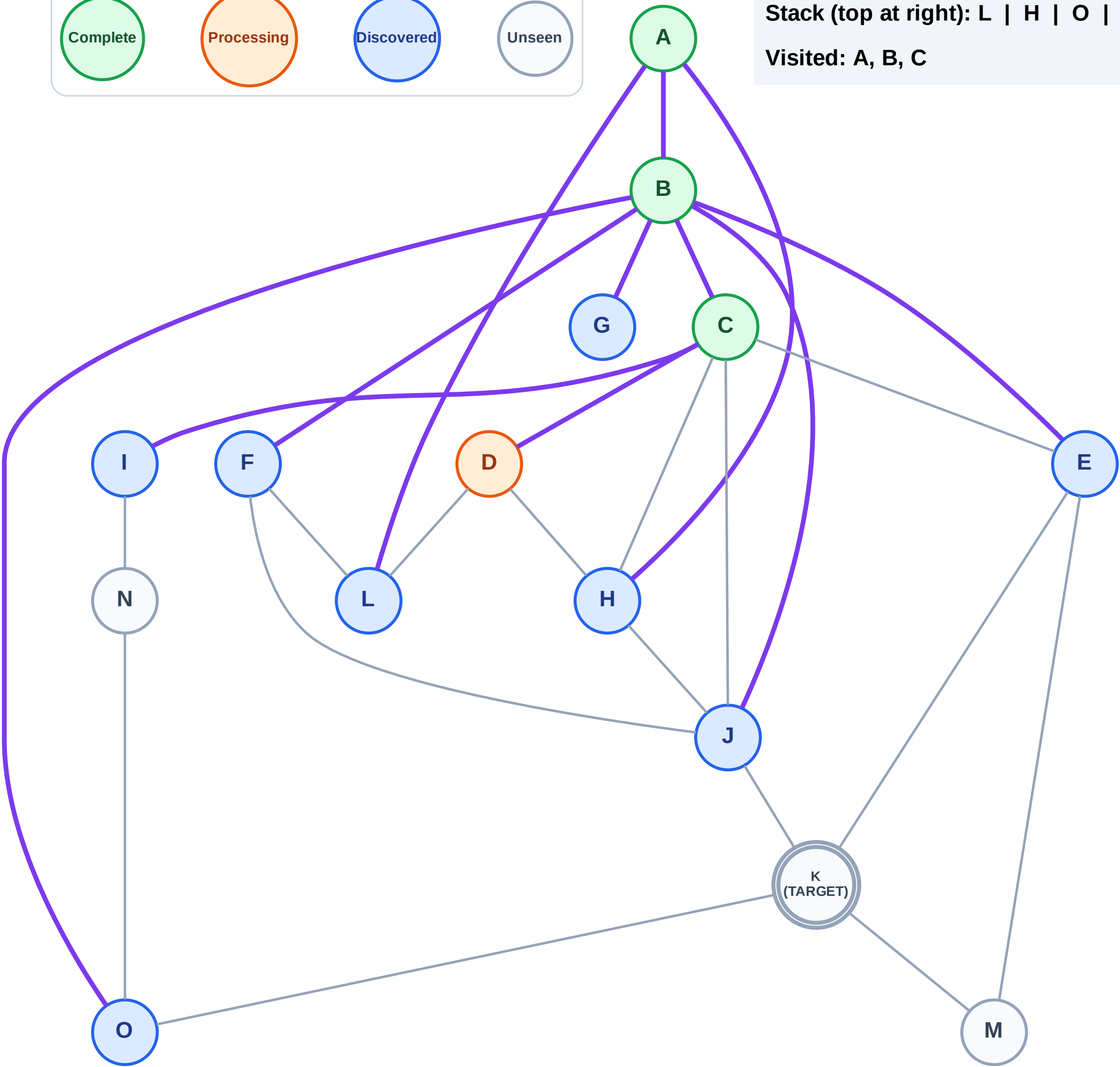
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E | I

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from D

Legend

Complete

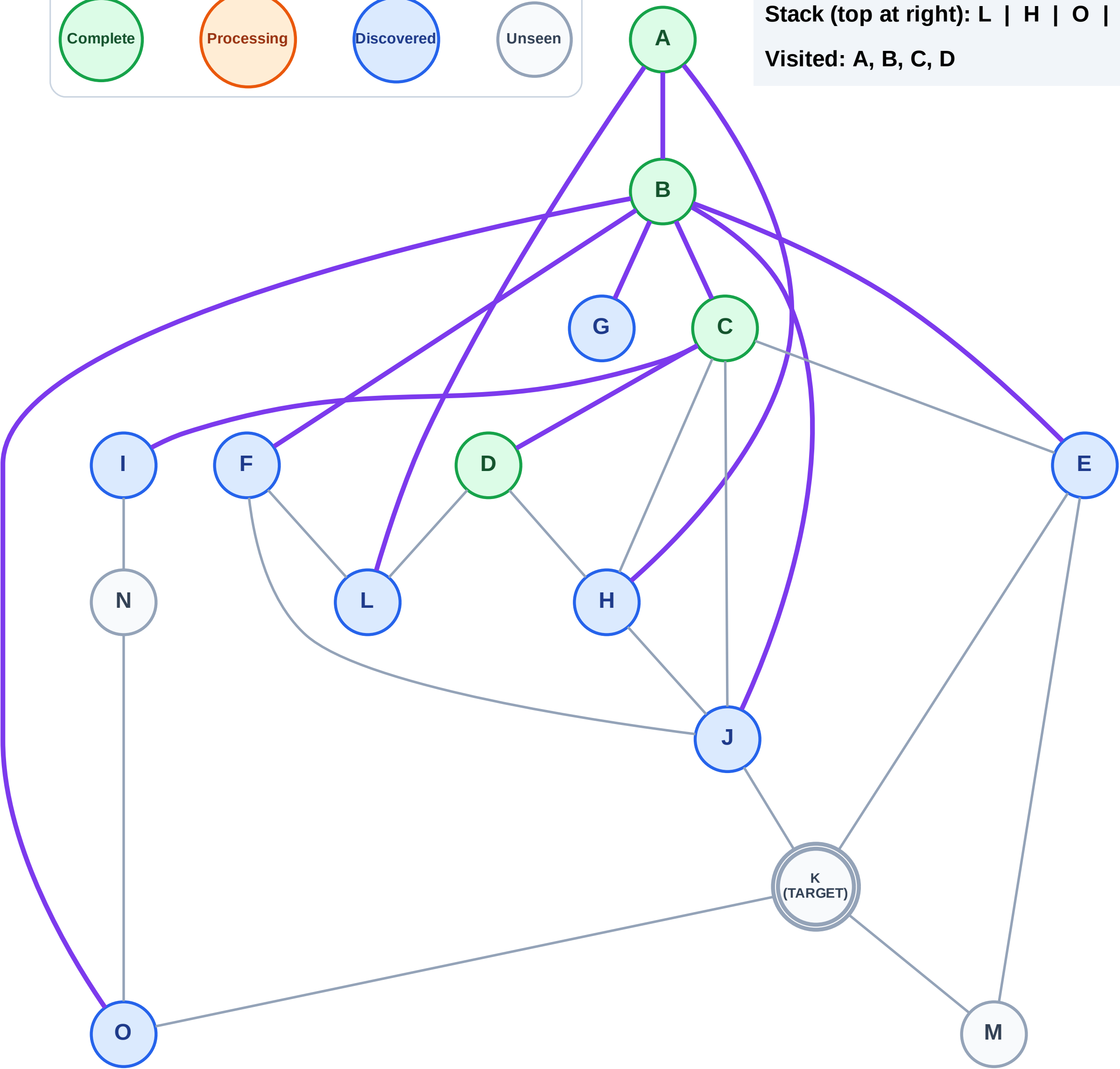
Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | E | I

Visited: A, B, C, D



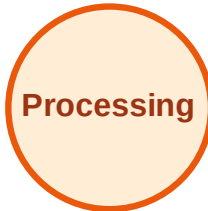
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



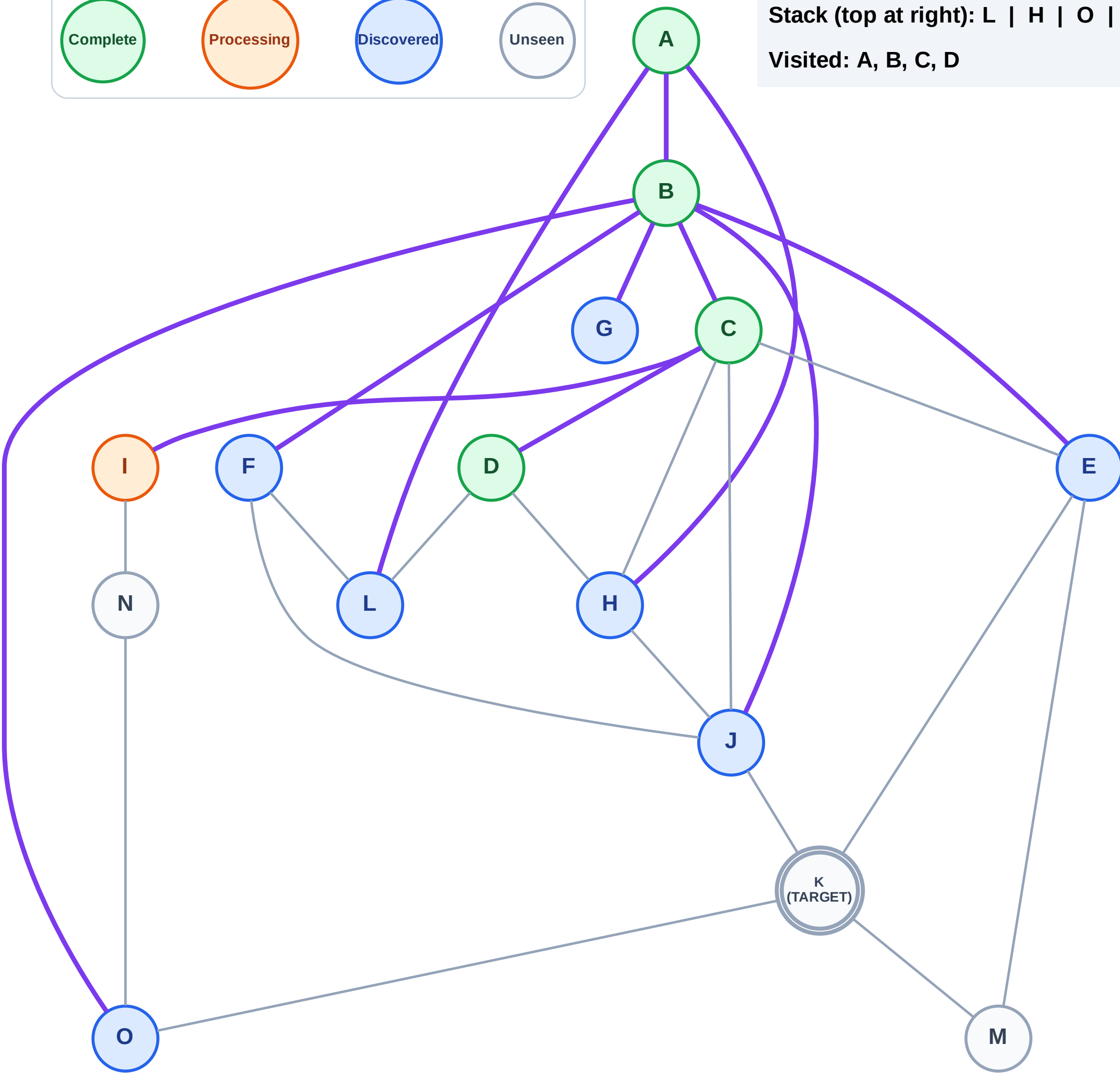
Discovered



Unseen

Stack (top at right): L | H | O | J | G | F | E

Visited: A, B, C, D



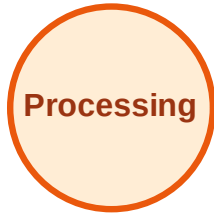
DFS Traversal

Step 11: Discover N from I

Legend



Complete



Processing



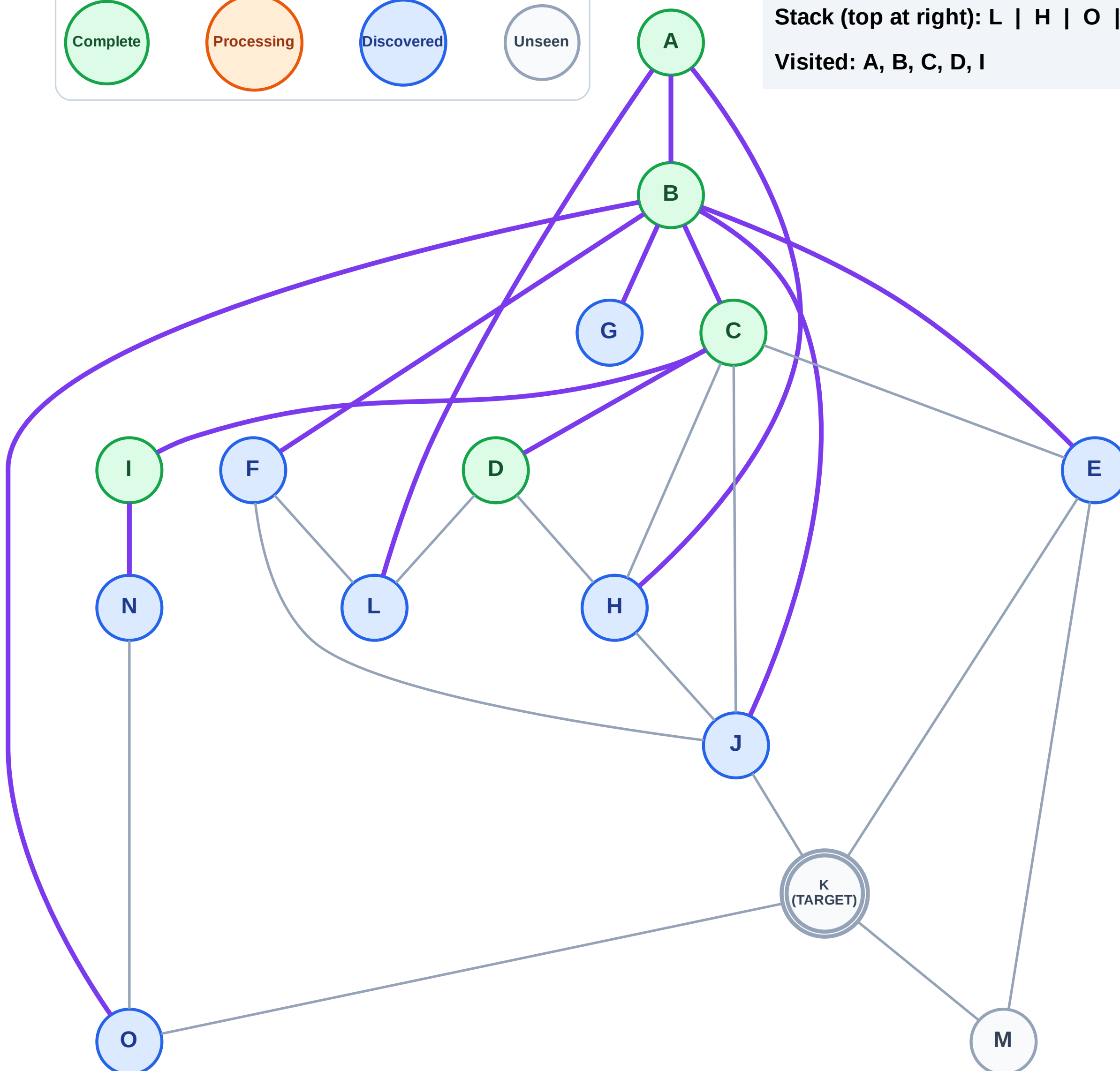
Discovered



Unseen

Stack (top at right): L | H | O | J | G | F | E | N

Visited: A, B, C, D, I



Step 12: Process node N

Step 12: Process node N



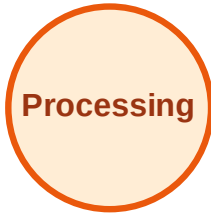
DFS Traversal

Step 13: No new neighbors from N

Legend



Complete



Processing



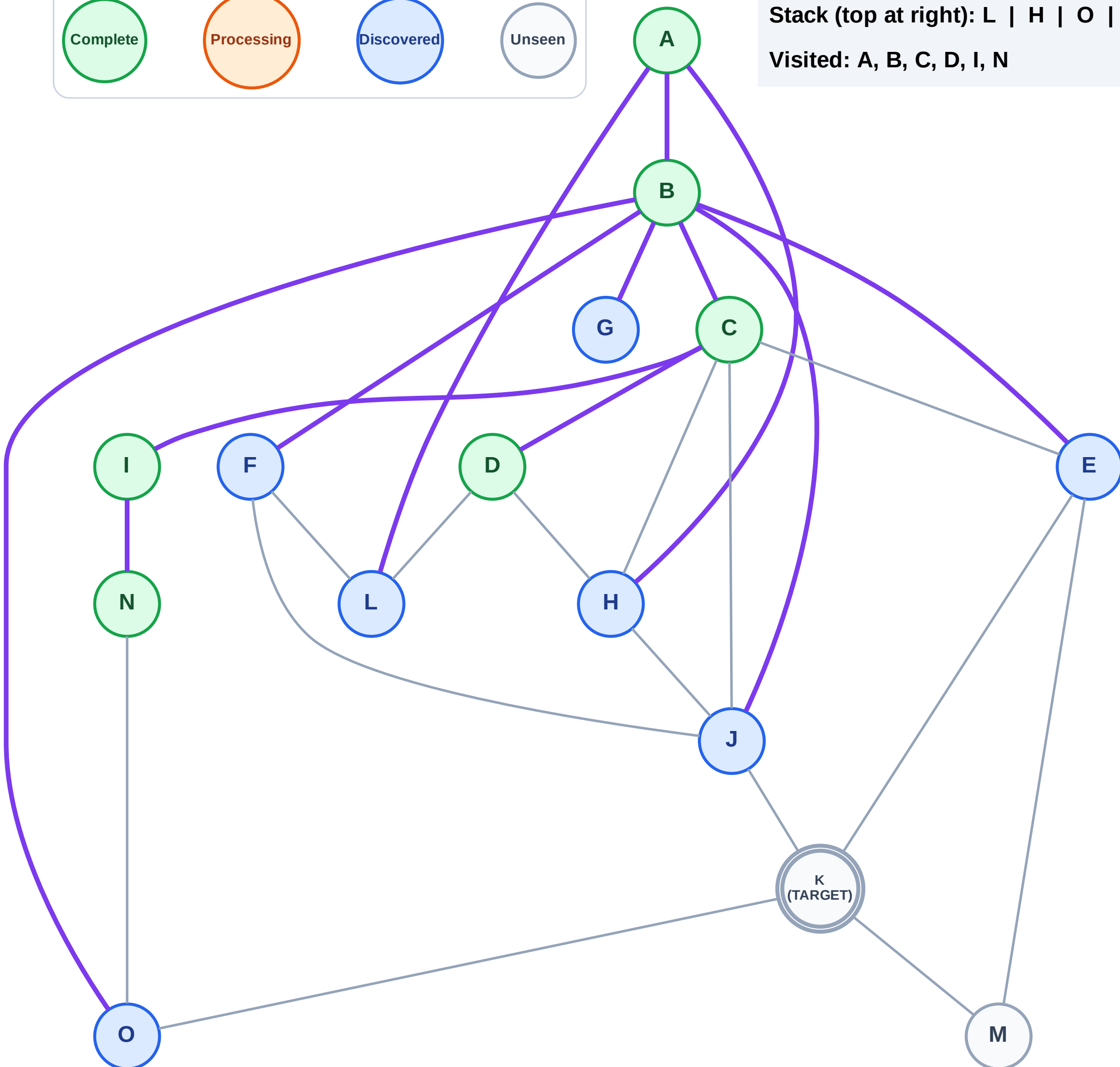
Discovered



Unseen

Stack (top at right): L | H | O | J | G | F | E

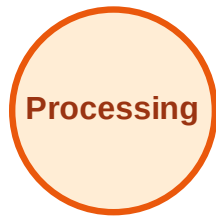
Visited: A, B, C, D, I, N



DFS Traversal

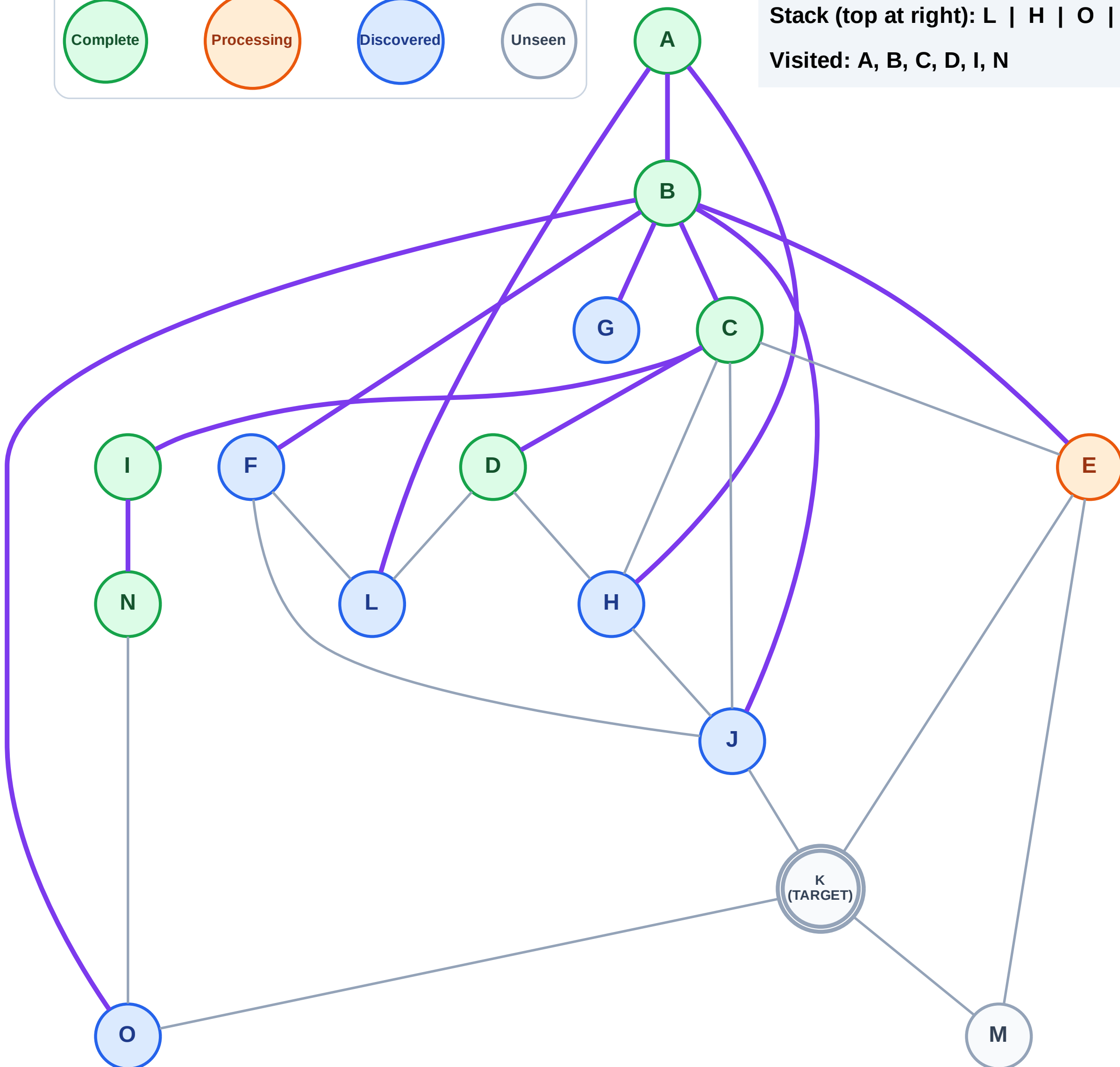
Step 14: Process node E

Legend



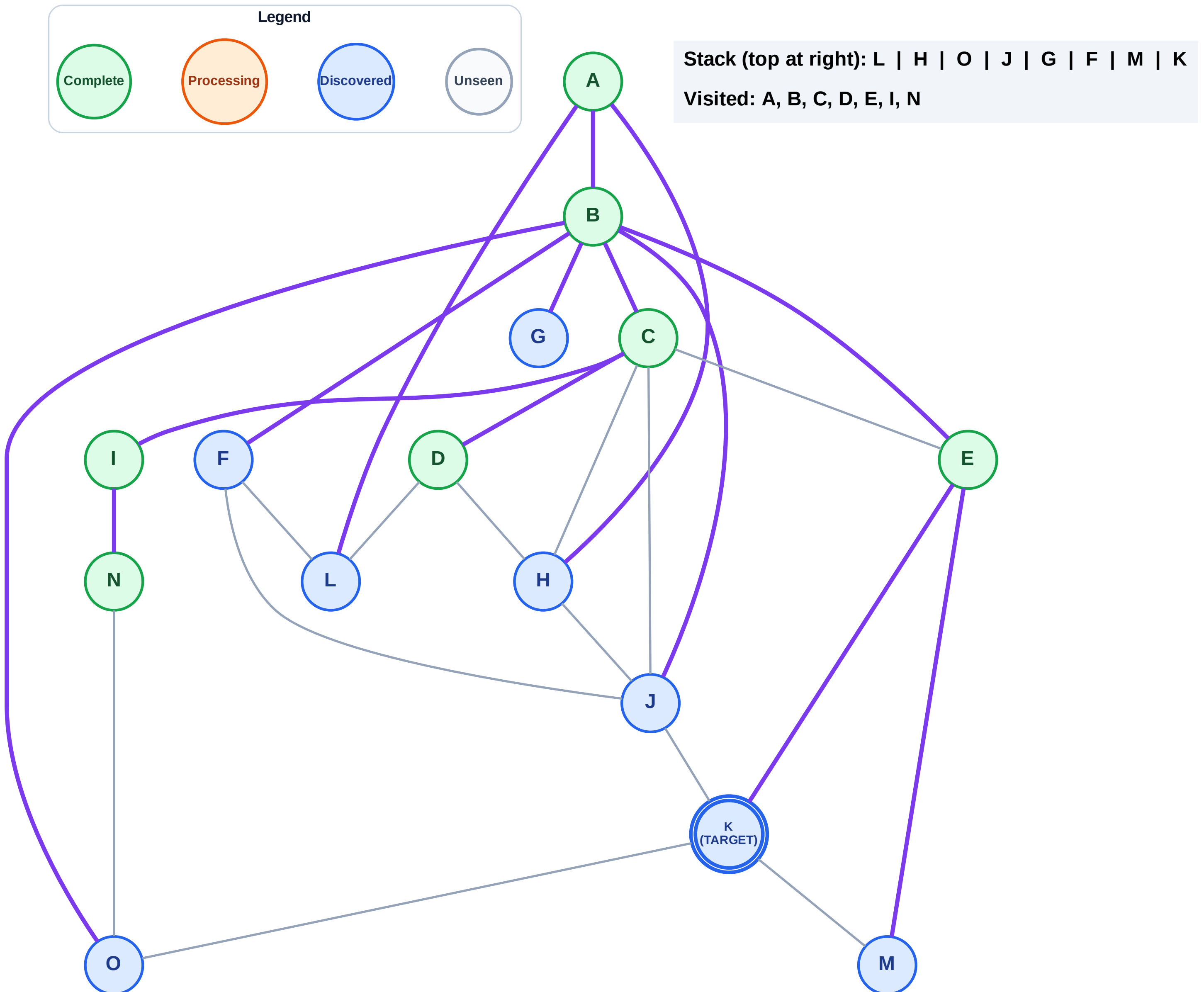
Stack (top at right): L | H | O | J | G | F

Visited: A, B, C, D, I, N



Step 15: Discover M, K from E

Step 15: Discover M, K from E



DFS Traversal

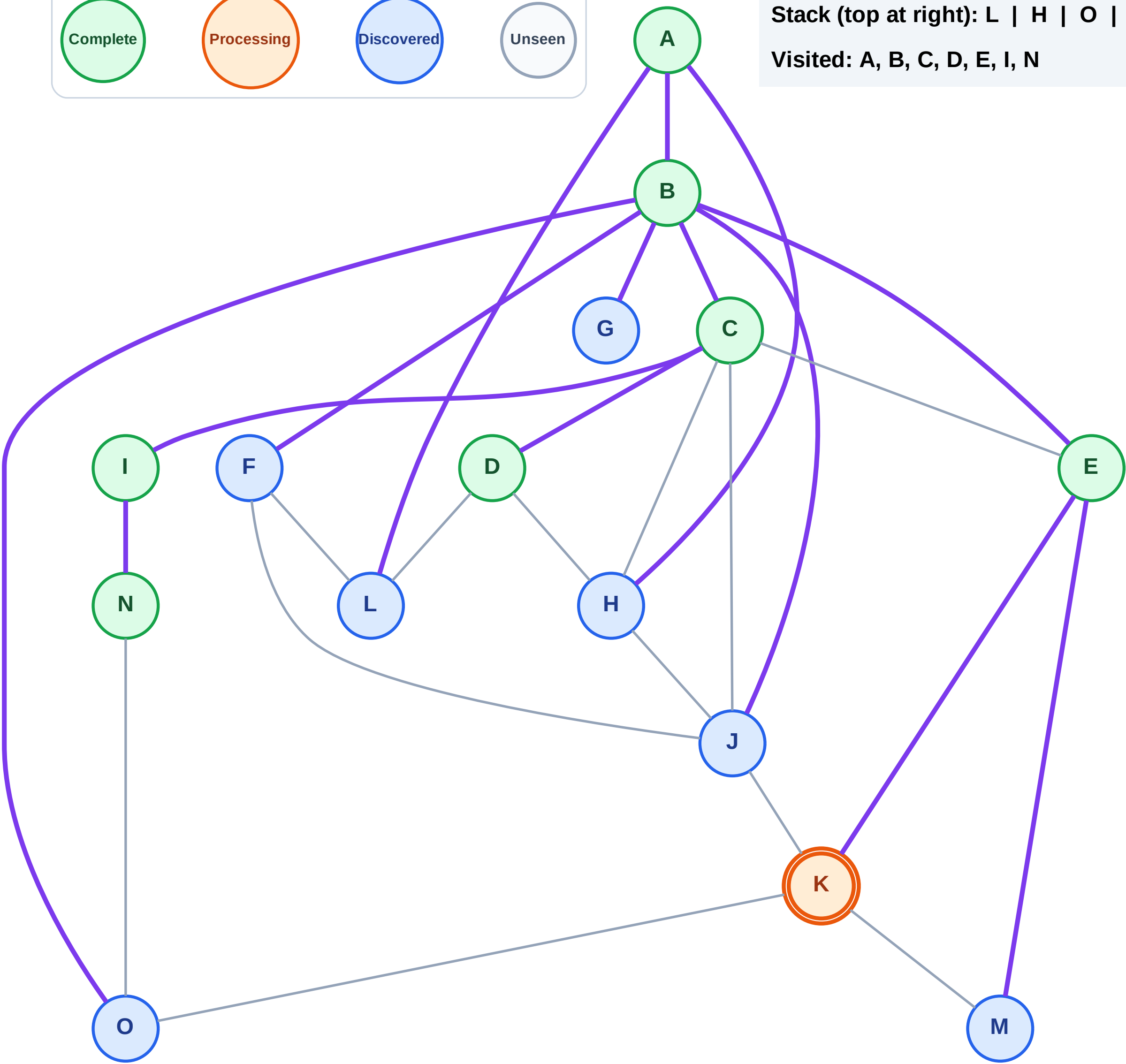
Step 16: Process node K

Legend



Stack (top at right): L | H | O | J | G | F | M

Visited: A, B, C, D, E, I, N



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | H | O | J | G | F | M

Visited: A, B, C, D, E, I, K, N

